

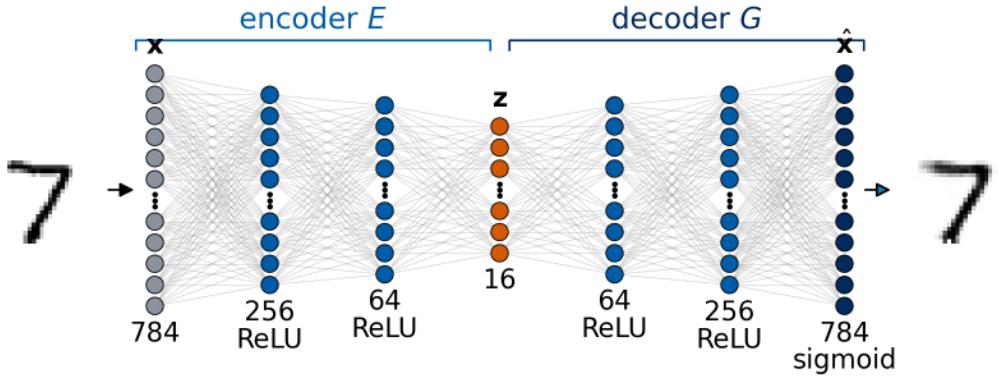
Lecture 6: Generative Models
Variational and Conditional Autoencoders
Generated Spins and the Hamiltonian

Deep Learning in Classical and Quantum Physics

Marcin Płodzień

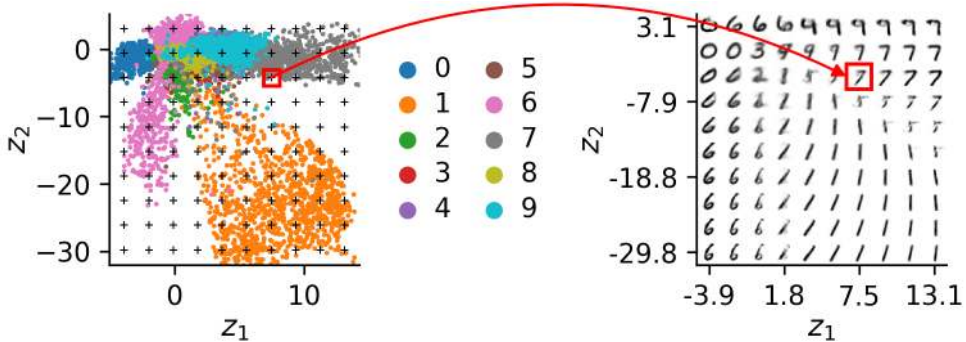
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The Autoencoder Compresses Each Image to a Point



Autoencoder (AE, Lecture 5): $\mathbf{z} = E(\mathbf{x}) \in \mathbb{R}^d$, $d \ll 784$, $\hat{\mathbf{x}} = G(\mathbf{z})$; trained with the mean squared error $\frac{1}{n}\sum_i \frac{1}{784}\|\mathbf{x}_i - G(E(\mathbf{x}_i))\|^2$. Each image becomes *one point*; nothing is learned about latent vectors no image was mapped to.

Decoding a Grid of Latent Points



AE, $d = 2$; left: test codes $E(\mathbf{x})$, + grid; right: **cell at $(z_1, z_2) = G(\mathbf{z})$** (grey = pixel value).
Grid: equal steps, 1st to 99th percentile of the codes; red: a point and its image.

- neighbouring cells decode to similar images (G is continuous)
- one region per digit; between two regions the image morphs into the other digit
- the empty space between islands of codes decodes to non-digits (G never trained there)

Random Latent Vectors Give No Digits



AE with $d = 16$, decoder applied to random $\mathbf{z}$. Top: $\mathbf{z} \sim \mathcal{N}(0, I)$; bottom: $z_k \sim \mathcal{N}(m_k, s_k^2)$, with m_k, s_k the mean and standard deviation of $E(\mathbf{x})_k$ over the training set.

- the codes lie on a thin, curved set in $\mathbb{R}^{16}$; a random point almost never falls on it, even with the right mean and spread per coordinate
- a convolutional neural network (CNN) classifier (Section 4) is confident, $\max_c p(c \mid \mathbf{x}) > 0.9$, for 37% of such images, against 96% of real test digits

To generate we need a known distribution on the latent space, and a decoder trained on all of it.

1

Learning a Probability Distribution



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Generative Models

The data $\mathbf{x}_1, \ldots, \mathbf{x}_n$ are **samples** from an unknown probability distribution $p(\mathbf{x})$: handwritten digits, or spin configurations at temperature T, $p(\mathbf{s} \mid T) = e^{-H(\mathbf{s})/T}/Z$.

A **generative model**

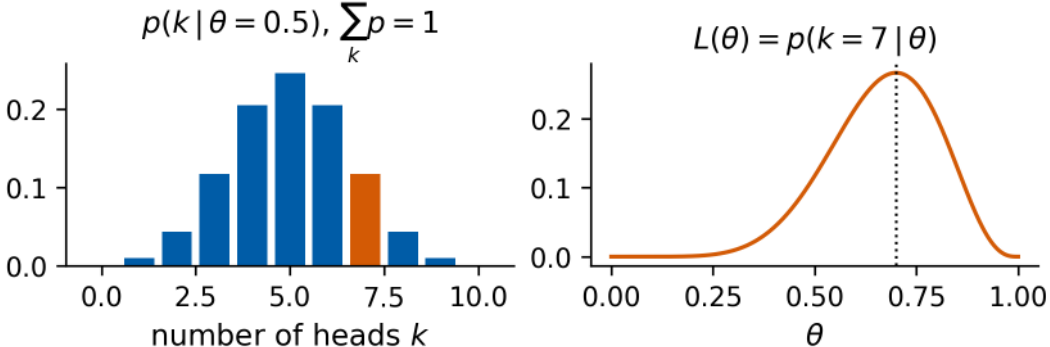
- learns an approximation $p_\theta(\mathbf{x}) \approx p(\mathbf{x})$ from the samples (θ: its parameters)
- draws **new samples** from it: instances that are not in the data set, but could have been

In science: new molecules and materials, ensembles of weather forecasts, fast surrogates of expensive simulations, and Monte Carlo samplers for lattice models.

Two steps: fit a distribution to data (maximum likelihood), and make it easy to sample (latent variables).

Probability and Likelihood

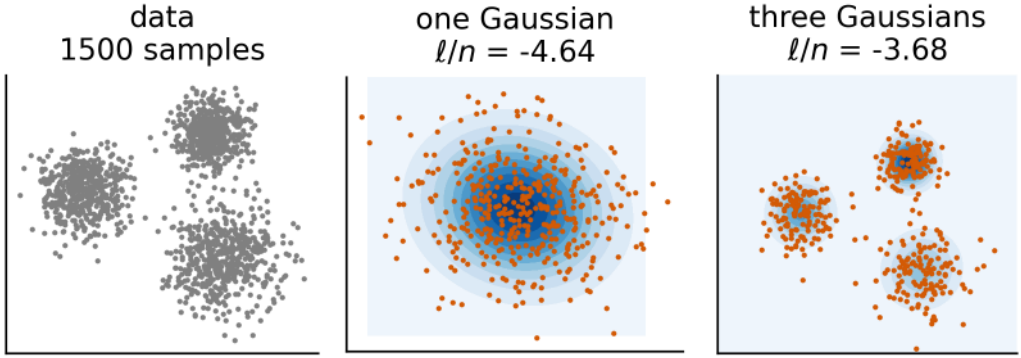
A model $p(x \mid \theta)$ is read in two ways: **probability** (θ fixed, x varies, $\sum_x p = 1$) and **likelihood** (data fixed, θ varies, $L(\theta) = p(x_{\text{obs}} \mid \theta)$, not a distribution over θ).



A coin tossed 10 times, $p(k \mid \theta) = \binom{10}{k}\theta^k(1-\theta)^{10-k}$. Left: $p(7 \mid 0.5) = 0.117$. Right: $k = 7$ observed, $\arg\max_\theta L = 0.7$, $\int_0^1 L\, d\theta = 1/11$.

Maximum Likelihood

Independent samples $\mathbf{x}_i$, model p_θ: $\ell(\theta) = \sum_{i=1}^{n} \log p_\theta(\mathbf{x}_i)$, $\theta^* = \arg\max_\theta \ell$. For $n \to \infty$, $\frac{1}{n}\ell \to -H(p) - \mathrm{KL}(p \,\|\, p_\theta)$, entropy $H(p) = -\mathbb{E}_p \log p$, **Kullback–Leibler divergence** $\mathrm{KL}(p\|p_\theta) = \mathbb{E}_p \log(p/p_\theta) \geq 0$ ($= 0$ iff $p_\theta = p$): $\max_\theta \ell \Leftrightarrow \min_\theta \mathrm{KL}(p \,\|\, p_\theta)$.



$p_\theta = \sum_k \pi_k \mathcal{N}(\mathbf{x}; \boldsymbol{\mu}_k, \Sigma_k)$, gradient ascent on ℓ; orange: 400 new samples.

A Mixture Is a Model with a Hidden Label

Read the fitted mixture as a two-step recipe: first a hidden label, then the point.

$$k \sim p(k) = \pi_k, \quad \mathbf{x} \sim p_\theta(\mathbf{x} \mid k) = \mathcal{N}(\mathbf{x}; \boldsymbol{\mu}_k, \Sigma_k) \quad \Longrightarrow \quad p_\theta(\mathbf{x}) = \sum_{k=1}^{K} p(k)\, p_\theta(\mathbf{x} \mid k)$$

- the label k is never observed: a **latent variable**; the orange points of the previous slide were drawn this way
- a digit is not one of a few prototypes (slant, thickness, size vary continuously): replace the discrete label by a **continuous latent vector** $\mathbf{z} \in \mathbb{R}^d$ and the sum by an integral; let a network turn $\mathbf{z}$ into the distribution of an image:

$$\mathbf{z} \sim p(\mathbf{z}), \qquad \mathbf{x} \sim p_\theta(\mathbf{x} \mid \mathbf{z}) \qquad \Longrightarrow \qquad p_\theta(\mathbf{x}) = \int p(\mathbf{z})\, p_\theta(\mathbf{x} \mid \mathbf{z})\, d\mathbf{z}$$

A mixture with one component for every z, shaped by a network: the variational autoencoder.

Four Distributions of a Latent-Variable Model

	in words	toy model: points near a curve
prior $p(\mathbf{z}) = \mathcal{N}(0, I)$	distribution of the hidden cause, fixed	position along the curve
decoder $p_\theta(\mathbf{x} \mid \mathbf{z})$	the observation once the cause is fixed; a network	small Gaussian around the curve point $\mathbf{g}_\theta(z)$
evidence $p_\theta(\mathbf{x}) = \int p(\mathbf{z})\, p_\theta(\mathbf{x}\|\mathbf{z})\, d\mathbf{z}$	the model of the data; what we sample	the spiral with its scatter
posterior $p_\theta(\mathbf{z} \mid \mathbf{x})$	which causes could have produced $\mathbf{x}$	narrow peak at the point's position

Bayes' rule: $p_\theta(\mathbf{z} \mid \mathbf{x}) = p_\theta(\mathbf{x} \mid \mathbf{z})\, p(\mathbf{z}) / p_\theta(\mathbf{x})$; the denominator is an integral over $\mathbb{R}^d$ we cannot compute $\Rightarrow$ replace the posterior by a network, the **encoder** $q_\phi(\mathbf{z} \mid \mathbf{x})$, a Gaussian with predicted mean and width.

A Latent Vector Is Like a Macrostate

Latent-variable model	Statistical mechanics
$\mathbf{z} \sim p(\mathbf{z})$: a few hidden numbers	macrostate, e.g. the magnetisation m
$\mathbf{x} \sim p_\theta(\mathbf{x}\|\mathbf{z})$: a full image, the remaining randomness spread over the pixels	microstate: all N spins, drawn with the bias set by m
$p_\theta(\mathbf{z}\|\mathbf{x})$: which hidden numbers produced this image	measuring the order parameter on a snapshot reads m off $\mathbf{s}$

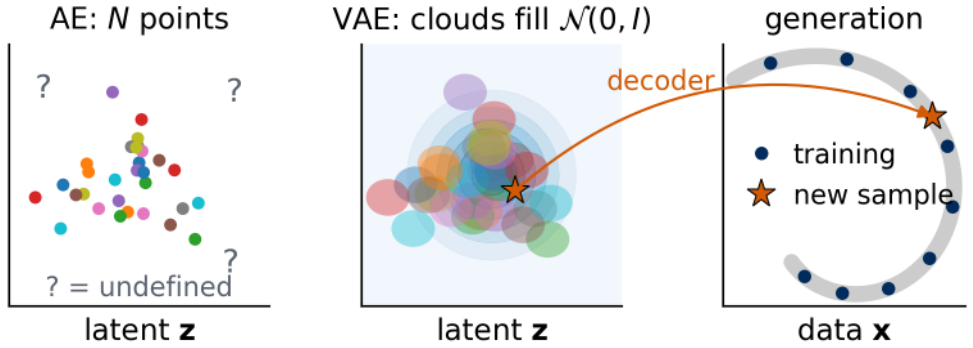
- a few macroscopic numbers fix the character of a configuration; generating = choose them, then fill in the details
- the posterior inverts the generation: it is what the encoder of a VAE approximates
- Lab 025 makes this literal: $\mathbf{x}$ = spin configuration, every spin drawn independently given $\mathbf{z}$ (Section 6)

2

Variational Autoencoders

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From a Discrete Set of Codes to a Continuous Distribution



- AE: N images $\rightarrow$ N latent *points*; between them the decoder is undefined
- wanted: a *continuous* density without holes, to interpolate and to draw new $\mathbf{z} \rightarrow$ new samples
- VAE: a cloud $q_\phi(\mathbf{z}|\mathbf{x})$ per image, all on $\mathcal{N}(0, I)$; new $\mathbf{z}$ (star) $\rightarrow$ decoder $\rightarrow$ new sample

Clouds Instead of Points

The AE failed for two reasons: no distribution on the latent space, and a decoder trained only on isolated codes. The **variational autoencoder** (VAE) changes what is trained:

- **prior** $p(\mathbf{z}) = \mathcal{N}(0, I_d)$, fixed: where latent vectors live
- **decoder** $p_\theta(\mathbf{x}\,|\,\mathbf{z}) = \prod_{j=1}^{784} \hat{x}_j^{\,x_j}(1-\hat{x}_j)^{1-x_j}$, $\hat{\mathbf{x}} = \sigma(G_\theta(\mathbf{z}))$: independent Bernoulli pixels, $\hat{x}_j =$ probability that pixel j is black
- **encoder** $q_\phi(\mathbf{z}\,|\,\mathbf{x}) = \mathcal{N}(\boldsymbol{\mu}_\phi(\mathbf{x}), \operatorname{diag}\boldsymbol{\sigma}^2_\phi(\mathbf{x}))$: a Gaussian *cloud* of latent vectors per image, approximating the posterior $p_\theta(\mathbf{z}|\mathbf{x})$

Training asks: points of the cloud of $\mathbf{x}$ decode back to $\mathbf{x}$ (decoder learns whole regions), and every cloud stays close to the prior (clouds of all images overlap and fill it).

D. P. Kingma, M. Welling, ICLR 2014, arXiv:1312.6114; D. J. Rezende, S. Mohamed, D. Wierstra, ICML 2014, PMLR 32, 1278

Generation: First z, then x

Generation after training: a sample of $p_\theta(\mathbf{x}) = \int p(\mathbf{z})\, p_\theta(\mathbf{x}|\mathbf{z})\, d\mathbf{z}$, drawn in two steps, first $\mathbf{z}$ from the prior, then $\mathbf{x}$ from the decoder; no encoder.

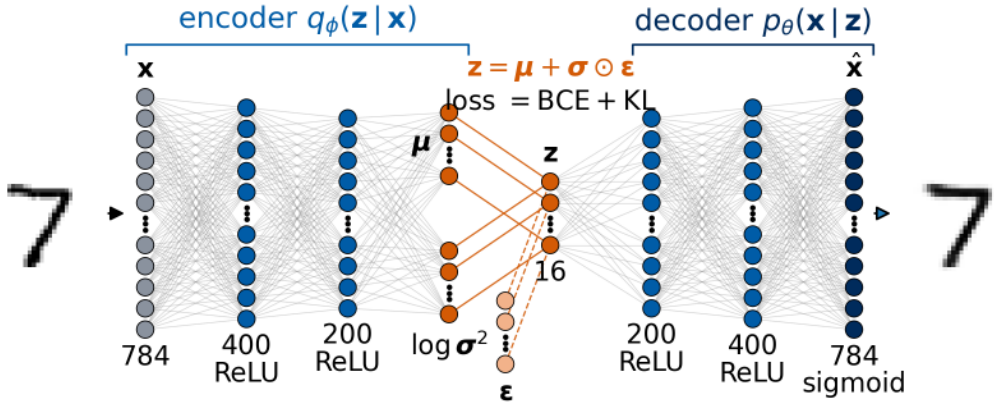
$\mathbf{z} \sim \mathcal{N}(0, I_d)$ → decoder G_θ → $\hat{\mathbf{x}} = \sigma(G_\theta(\mathbf{z}))$ → $x_j \sim \text{Bernoulli}(\hat{x}_j)$, or show $\hat{\mathbf{x}}$

Training needs $\log p_\theta(\mathbf{x})$, an integral over $\mathbb{R}^d$. Two estimates with K samples $\mathbf{z}_k$:

$$\underbrace{\log \frac{1}{K} \sum_{k=1}^{K} p_\theta(\mathbf{x}|\mathbf{z}_k)}_{\mathbf{z}_k \sim \mathcal{N}(0,I)} \qquad \text{or} \qquad \underbrace{\log \frac{1}{K} \sum_{k=1}^{K} \frac{p_\theta(\mathbf{x}|\mathbf{z}_k)\, p(\mathbf{z}_k)}{q_\phi(\mathbf{z}_k|\mathbf{x})}}_{\mathbf{z}_k \sim q_\phi(\mathbf{z}|\mathbf{x}) \text{ (importance sampling)}}$$

$d = 16$, $K = 1000$, mean over 200 test digits: prior -154.8 nats, encoder -93.3 nats. Almost no prior draw lands where $p_\theta(\mathbf{x}|\mathbf{z})$ is large: sampling the prior fails, the encoder is needed.

The VAE Network: Two Heads and a Sampling Node



$\varepsilon \sim \mathcal{N}(0, I)$: a separate input without parameters; $z_j = \mu_j + \sigma_j \varepsilon_j$ element-wise (orange lines).

The Evidence Lower Bound

Kullback–Leibler divergence: $\mathrm{KL}(q\|p) = \mathbb{E}_q\big[\log \frac{q}{p}\big] \geq 0$, $= 0$ only for $q = p$; the average log-ratio over samples of q: how distinguishable q is from p, in nats.
Bayes' rule for every $\mathbf{z}$, $\log p_\theta(\mathbf{x}) = \log p_\theta(\mathbf{x}|\mathbf{z}) + \log p(\mathbf{z}) - \log p_\theta(\mathbf{z}|\mathbf{x})$; insert $q_\phi(\mathbf{z}|\mathbf{x})$ and average over it:

$$\log p_\theta(\mathbf{x}) = \underbrace{\mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x})}\big[\log p_\theta(\mathbf{x}\,|\,\mathbf{z})\big] - \mathrm{KL}\big(q_\phi(\mathbf{z}|\mathbf{x}) \,\|\, p(\mathbf{z})\big)}_{\mathrm{ELBO}(\theta,\phi;\,\mathbf{x})} + \underbrace{\mathrm{KL}\big(q_\phi(\mathbf{z}|\mathbf{x}) \,\|\, p_\theta(\mathbf{z}|\mathbf{x})\big)}_{\geq 0}$$

- **ELBO** = evidence lower bound: a *computable* lower bound on the log-likelihood (the "evidence") $\log p_\theta(\mathbf{x})$; nats per image (random pixels: $784 \ln \frac{1}{2} = -543$)
- tight when q_ϕ equals the true posterior: maximising the ELBO raises $\log p_\theta(\mathbf{x})$ and pulls q_ϕ towards $p_\theta(\mathbf{z}|\mathbf{x})$
- trained $d = 16$ VAE: ELBO $= -98.1$, $\log p_\theta(\mathbf{x}) \approx -93.3$ nats

$\log p_\theta(\mathbf{x})$ by importance sampling: Y. Burda, R. Grosse, R. Salakhutdinov, arXiv:1509.00519

The VAE Loss: Reconstruction and the Price of Information

Loss per image $= -\text{ELBO}$, one sample $\mathbf{z} \sim q_\phi(\mathbf{z}|\mathbf{x})$:

$$\mathcal{L}(\mathbf{x}) = \underbrace{\mathrm{BCE}(\mathbf{x}, \hat{\mathbf{x}})}_{-\log p_\theta(\mathbf{x}|\mathbf{z})} + \underbrace{\frac{1}{2}\sum_{k=1}^{d} (\mu_k^2 + \sigma_k^2 - \ln \sigma_k^2 - 1)}_{\mathrm{KL}(q_\phi(\mathbf{z}|\mathbf{x}) \,\|\, \mathcal{N}(0,I))}, \qquad \hat{\mathbf{x}} = \sigma(G_\theta(\mathbf{z}))$$

- **binary cross-entropy** $\mathrm{BCE}(\mathbf{x}, \hat{\mathbf{x}}) = -\sum_j [x_j \ln \hat{x}_j + (1 - x_j) \ln(1 - \hat{x}_j)]$: negative log-likelihood of independent Bernoulli pixels with means $\hat{x}_j$; per pixel 0 if certain and correct, $\ln 2 = 0.69$ nats for $\hat{x}_j = \frac{1}{2}$; alone, an autoencoder (points, not clouds)
- **KL term**: zero only for $\boldsymbol{\mu} = 0$, $\boldsymbol{\sigma} = 1$, a cloud that says nothing about $\mathbf{x}$; moving clouds apart (μ_k^2) and narrowing them ($-\ln \sigma_k^2$) costs nats: the price of the information $\mathbf{z}$ carries about $\mathbf{x}$
- clouds pulled onto $\mathcal{N}(0, I)$ cover the prior: a fresh $\mathbf{z} \sim \mathcal{N}(0, I)$ lands where the decoder was trained

The Negative ELBO Is a Variational Free Energy

States $\mathbf{z}$, energy $E(\mathbf{z})$, temperature T: Boltzmann distribution $e^{-E/T}/Z$, free energy $F = -T \log Z$. For a trial distribution q: $F[q] = \langle E \rangle_q - TS[q]$, entropy $S[q] = -\mathbb{E}_q \log q$.

For one image, take $E(\mathbf{z}) = -\log[p_\theta(\mathbf{x}|\mathbf{z})\, p(\mathbf{z})]$ and $T = 1$:

$$\frac{e^{-E}}{Z} = p_\theta(\mathbf{z}|\mathbf{x}), \qquad Z = p_\theta(\mathbf{x}), \qquad F = -\log p_\theta(\mathbf{x})$$

$$F[q] = F + \mathrm{KL}(q \,\|\, e^{-E}/Z) \geq F \quad \text{(Gibbs–Bogoliubov)}, \qquad F[q] = -\mathrm{ELBO}$$

- training a VAE minimises a variational free energy; the encoder is a Gaussian trial distribution, one per image
- the same inequality underlies mean-field theory: the best product distribution bounds F from above

The Reparametrisation Trick

$\mathbb{E}_q$ is estimated with one sample of $\mathbf{z}$ per image. But "draw $\mathbf{z}$ from $\mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\sigma}^2)$" has no derivative with respect to $\boldsymbol{\mu}$ and $\boldsymbol{\sigma}$.

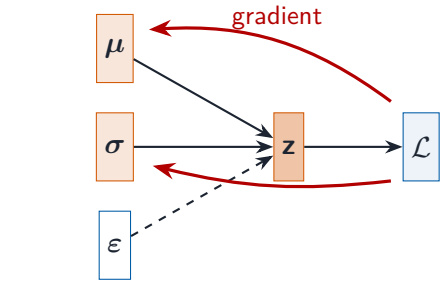
Move the randomness out of the network:

$$\mathbf{z} = \boldsymbol{\mu} + \boldsymbol{\sigma} \odot \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, I), \quad \boldsymbol{\sigma} = e^{\frac{1}{2}\log \boldsymbol{\sigma}^2}$$

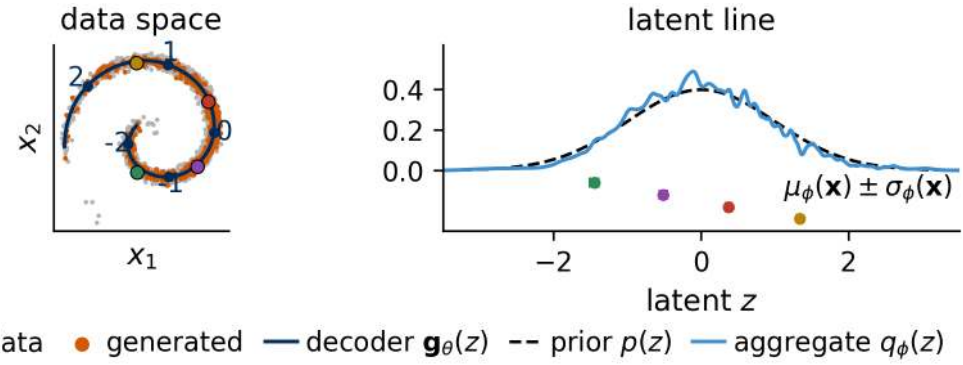
$$\frac{\partial z_k}{\partial \mu_k} = 1, \qquad \frac{\partial z_k}{\partial \sigma_k} = \varepsilon_k$$

ε does not depend on the parameters; backpropagation runs from the loss through decoder, $\mathbf{z}$ and encoder.

In JAX, ε comes from an explicit key: `eps = jax.random.normal(key, mu.shape)`.

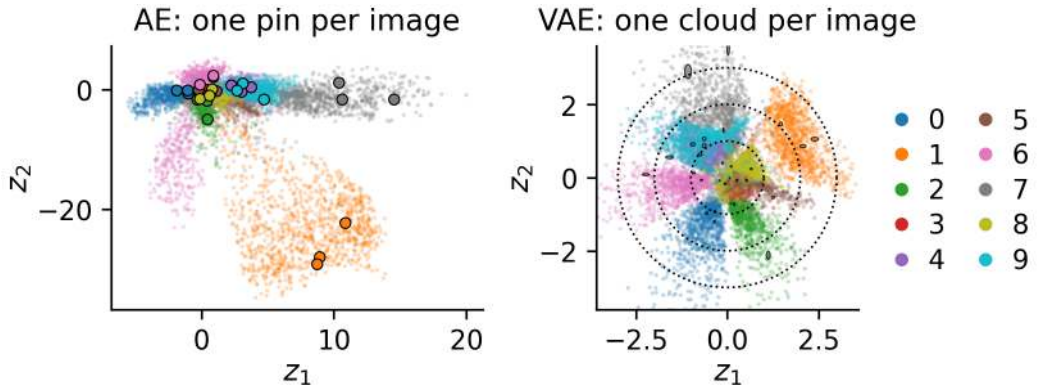


A VAE on Points near a Curve



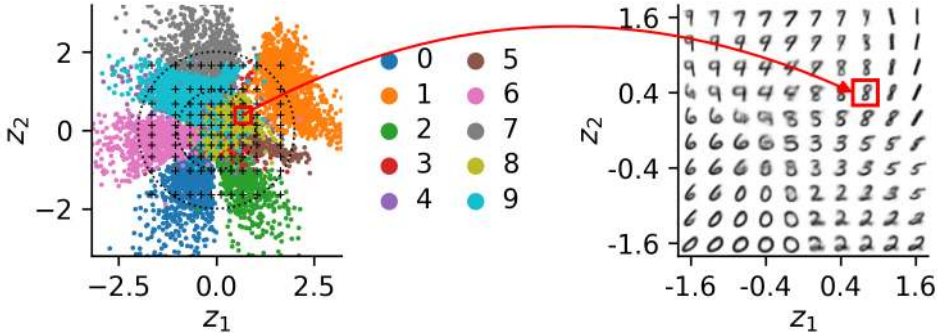
$\mathbf{x} = \mathbf{f}(z_{\text{true}}) + 0.08\,\boldsymbol{\eta}$ on a spiral; VAE with 1D latent, $p_\theta(\mathbf{x}|z) = \mathcal{N}(\mathbf{g}_\theta(z), 0.08^2 I)$. Dots right: clouds $\mu_\phi \pm \sigma_\phi$ of the coloured points (median $\sigma_\phi = 0.04$); **aggregate posterior** $q_\phi(z) = \frac{1}{n}\sum_i q_\phi(z|\mathbf{x}_i) \approx p(z)$; $|\mathrm{corr}(\mu_\phi, z_{\text{true}})| = 0.993$.

The KL Term Turns Islands into One Covered Disc



$d = 2$; dots: all test digits; pins / ellipses $\boldsymbol{\mu}_\phi \pm \boldsymbol{\sigma}_\phi$: three per class; dotted: $|\mathbf{z}| = 1, 2, 3$. Coordinate std: AE 3.8, 7.5, VAE 1.03, 1.09; median $\boldsymbol{\sigma}_\phi = 0.04, 0.04$. AE = a pin per image, islands in an empty sea; VAE = a cloud per image, pulled onto one disc with no sea: **sampling = throwing a dart at the disc**.

Every Grid Point of the VAE Decodes to a Digit



VAE, $d = 2$, layout as for the AE; left: means $\boldsymbol{\mu}_\phi(\mathbf{x})$. **Cell at (z_1, z_2) = decoder mean** $\sigma(G_\theta(\mathbf{z}))$ (grey = pixel probability). Grid: $z^{(j)} = \Phi^{-1}\big((j + \frac{1}{2})/10\big)$, Φ = normal CDF: each cell has prior mass $1/100$.

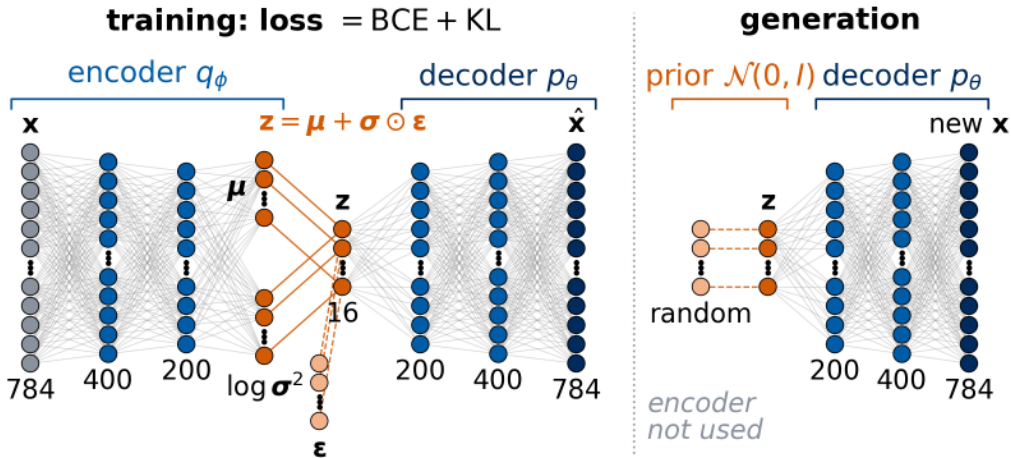
- clouds cover the prior disc: no nonsense cells; one region per digit, morphing at borders
- blurry: the mean averages all pixel patterns compatible with $\mathbf{z}$

3

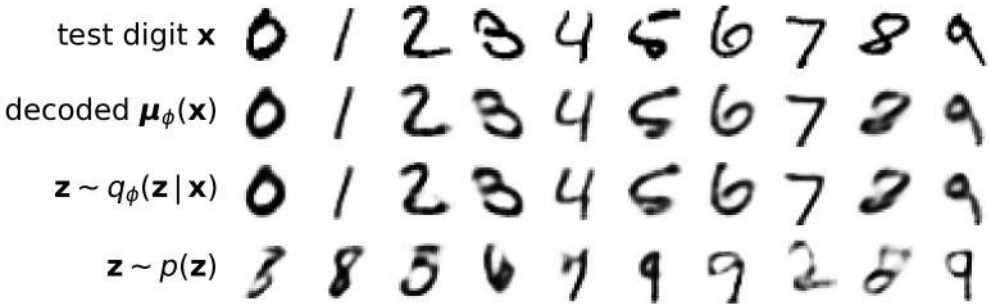
Sampling a Trained VAE

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Training and Generation Use Different Parts of the VAE

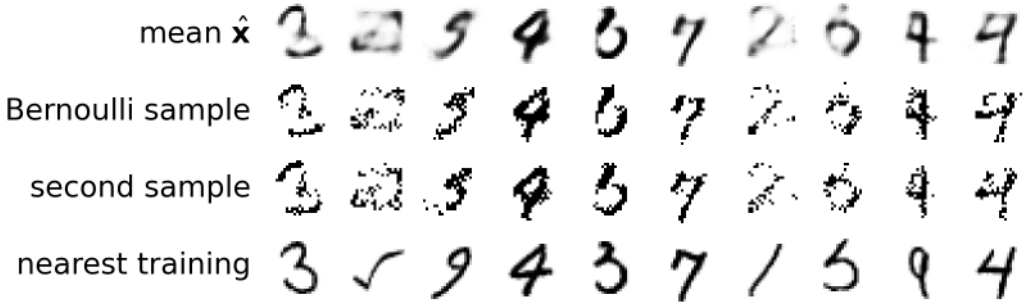


Reconstruction and Generation



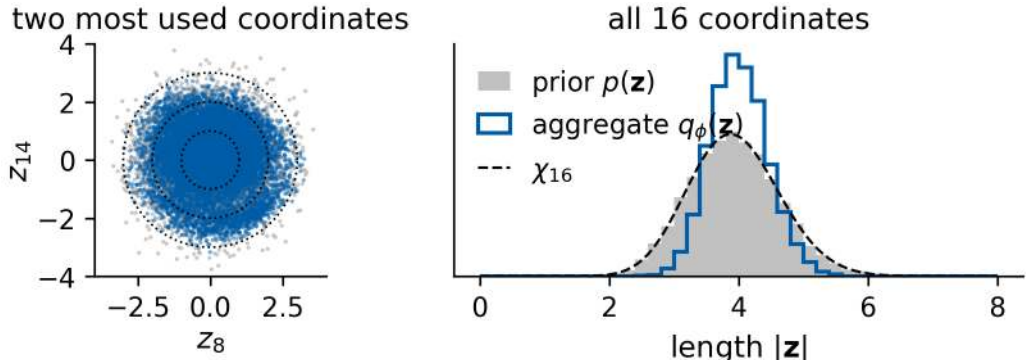
- **reconstruction** $\mathbf{x} \to \mathbf{z} \sim q_\phi(\mathbf{z}|\mathbf{x}) \to p_\theta(\mathbf{x}|\mathbf{z})$: identity and most of the style kept
- **generation** $\mathbf{z} \sim p(\mathbf{z}) \to p_\theta(\mathbf{x}|\mathbf{z})$: only decoder and prior are kept
- $d = 16$, decoder means; confidence > 0.9 of a classifier (Section 4): real 96%, reconstr. 92%, generated 54%

A Sample Is Not the Decoder Mean



- shown so far: the mean $\hat{x}_j$ (blurry); a **sample** draws each pixel (black with prob. $\hat{x}_j$): speckled
- correlations (a continuous stroke) pass through $\mathbf{z}$; the rest are coin flips, as for spins (Section 6)
- not copies: distance to the nearest training image 4.48 (generated), 4.19 (unseen test)

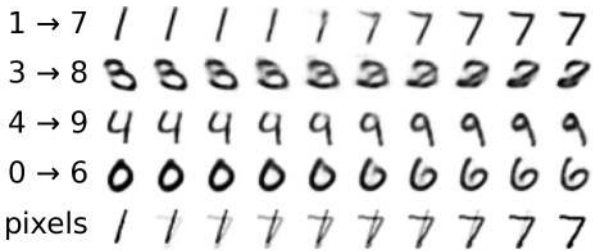
The Encoded Data Do Not Fill the Prior Exactly



Aggregate posterior $q_\phi(\mathbf{z}) = \frac{1}{n}\sum_i q_\phi(\mathbf{z}|\mathbf{x}_i)$: where the encoded data live (sampled: encode a test image, draw from its cloud). Grey: $\mathcal{N}(0, I)$; χ_{16}: distribution of $|\mathbf{z}|$ under the prior.

- per-coordinate std of $q_\phi(\mathbf{z})$ between 0.97 and 1.04; mean $|\mathbf{z}|$: 4.01 (q_ϕ), 3.93 (prior)
- median distance to the nearest test code: prior samples 2.61, other codes 1.55: **prior holes**

Interpolation in the Latent Space

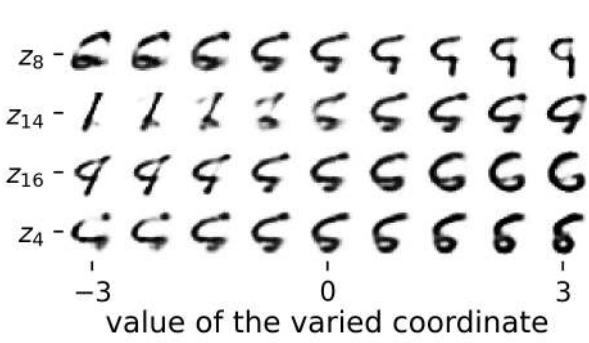


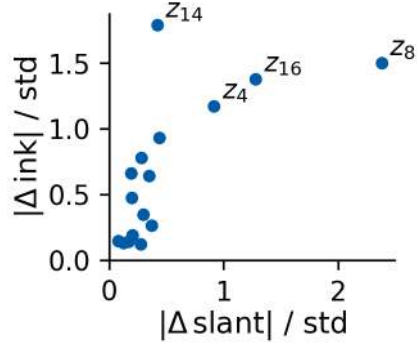
$\mathbf{z}(t) = (1-t)\,\boldsymbol{\mu}_\phi(\mathbf{x}_a) + t\,\boldsymbol{\mu}_\phi(\mathbf{x}_b)$, $t \in [0,1]$, decoded by the $d = 16$ VAE; last row: $(1-t)\,\mathbf{x}_a + t\,\mathbf{x}_b$.

- latent line: plausible digits throughout; the 1 bends into a 7, the 4 closes into a 9
- pixel line: two superposed digits

Straight lines in the latent space stay among the data.

One Latent Coordinate Changes Several Properties at Once





Left: $\boldsymbol{\mu}_\phi$ of one test 5, one coordinate varied from -3 to 3 (four most used, by $\overline{\mathrm{KL}_k} = \langle \mathrm{KL} \text{ of coordinate } k \rangle_{\mathbf{x}}$, from 0.65 to 2.23 nats). Right: change of slant $-\mathrm{Cov}(u, v)/\mathrm{Var}(v)$ and of ink (mean pixel value) for $z_k = -2 \to 2$, 300 test digits, in units of their spread over real digits; one dot per coordinate. 7 of 16 coordinates change both by > 0.3.

The Goal: One Latent Coordinate per Property

- the ELBO does not choose the axes: $\mathcal{N}(0, I)$ and the KL term are unchanged by a rotation $\mathbf{z} \to R\mathbf{z}$, and the decoder absorbs it, $G'(\mathbf{z}) = G(R^\top \mathbf{z})$; a rotated latent space, every axis mixing slant, thickness and identity, is as good as any other
- **disentangled** code: each coordinate carries one distinct, independent factor of variation (slant, thickness, size); it can be read, controlled, and in physics it can *discover* a relevant variable (a coordinate = the magnetisation)
- without labels or further assumptions not guaranteed; it can be encouraged

Factor unknown: encourage independence

β-VAE: KL term $\times\, \beta > 1$, information in $\mathbf{z}$ more expensive, stronger pull to the factorised prior $\prod_k \mathcal{N}(0,1)$ (Lab 026)

Factor known and labelled: give it as input

conditional VAE: the label enters encoder and decoder, $\mathbf{z}$ keeps the rest (Section 5)

F. Locatello et al., *Challenging Common Assumptions in the Unsupervised Learning of Disentangled Representations*, ICML 2019, PMLR 97, 4114

4

Judging Synthetic Data



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A Classifier as a Referee

Pictures are judged by eye. To *measure* what a generator produces, use a referee trained on the **real** data, with known accuracy on real test data, and apply it unchanged to synthetic images.

- CNN: 3×3 convolutions with 16 and 32 channels, ReLU, each followed by 2×2 max-pooling ($28 \to 14 \to 7$); dense 128; 10 outputs $f_c(\mathbf{x})$
- trained with the cross-entropy, the negative log-likelihood of the labels (Adam, 4 epochs):

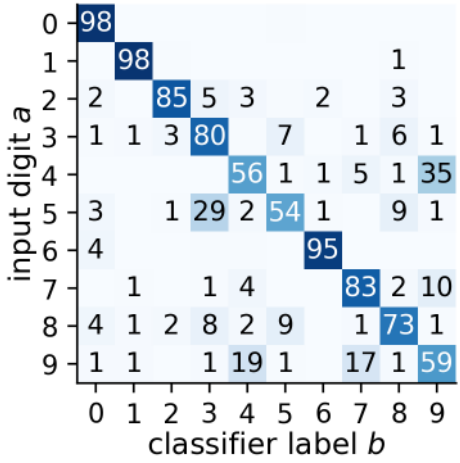
$$\mathcal{L} = -\frac{1}{n} \sum_i \log p(y_i \,|\, \mathbf{x}_i), \qquad p(c \,|\, \mathbf{x}) = \frac{e^{f_c(\mathbf{x})}}{\sum_{c'} e^{f_{c'}(\mathbf{x})}}$$

test accuracy on 10 000 real digits

0.9854

Every test of synthetic data is first run on real data: the baseline.

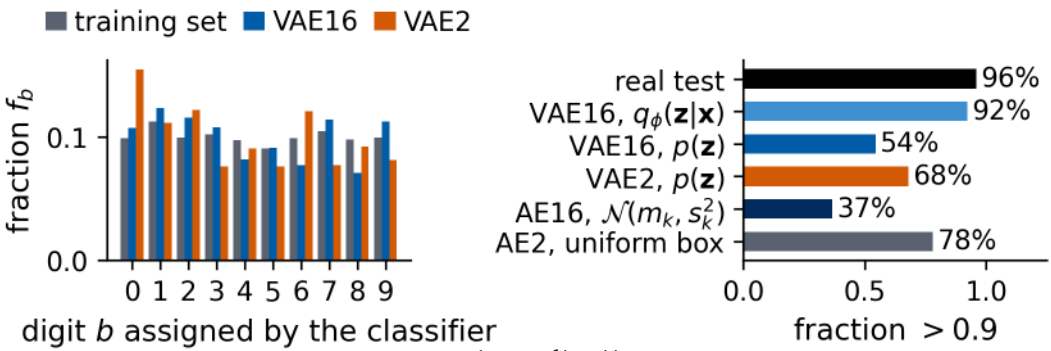
Confusion Matrix of VAE Reconstructions



Row a: digit of the real test input $\mathbf{x}$; column b: classifier label of the reconstruction $\sigma(G_\theta(\mathbf{z}))$, $\mathbf{z} \sim q_\phi(\mathbf{z} \mid \mathbf{x})$; C_{ab} in % of row a. Shown: $d = 2$.

- mean diagonal $\frac{1}{10}\mathrm{tr}\, C$: real images 0.985, $d = 16$ 0.953, $d = 2$ 0.783
- $d = 16$: the digit survives the latent space
- $d = 2$: look-alike digits share a region of the plane; worst: $4 \to 9$: 35%

Digit Distribution of the Generated Samples



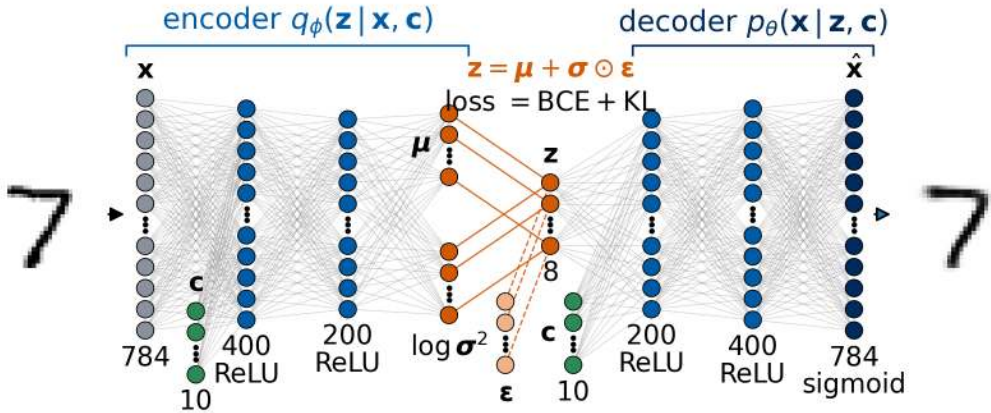
Left: f_b = fraction of $N = 10\,000$ samples ($\mathbf{z} \sim \mathcal{N}(0, I)$) assigned to b. Right: fraction with confidence $\max_c p(c \mid \mathbf{x}) > 0.9$; VAE16 q_ϕ: reconstructions.

- $d = 16$: training frequencies (no mode collapse); $d = 2$: too many 0s, too few 3s, 5s, 7s
- reconstructions clearer than generated samples: prior holes (Section 3)

5
Conditional Generation

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The Conditional VAE Receives the Label as an Input



Known, labelled factor $\Rightarrow$ give it to both networks: conditional VAE (cVAE); $\mathbf{c}$ = one-hot digit, $c_j = \delta_{j,y}$ (a single number would put 3 "between" 2 and 4); $d = 8$.

The Conditioned ELBO

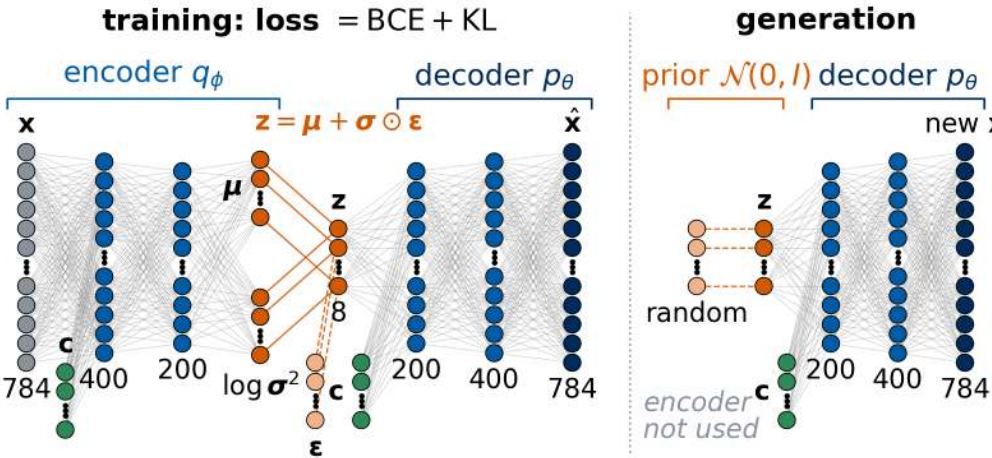
K. Sohn, H. Lee, X. Yan, *Learning Structured Output Representation using Deep Conditional Generative Models*, NeurIPS 2015

The same derivation with $\mathbf{c}$ carried along gives a lower bound on the conditional likelihood:

$$\log p_\theta(\mathbf{x}\,|\,\mathbf{c}) \;\geq\; \mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x},\mathbf{c})}\big[\log p_\theta(\mathbf{x}\,|\,\mathbf{z},\mathbf{c})\big] - \mathrm{KL}\big(q_\phi(\mathbf{z}\,|\,\mathbf{x},\mathbf{c})\,\big\|\,p(\mathbf{z})\big), \qquad p(\mathbf{z}) = \mathcal{N}(0, I)$$

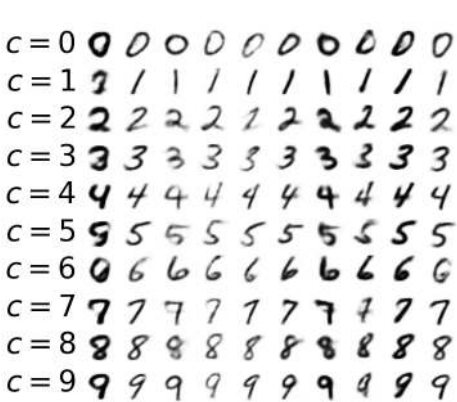
- what we train: encoder $q_\phi(\mathbf{z}|\mathbf{x},\mathbf{c})$, decoder $p_\theta(\mathbf{x}|\mathbf{z},\mathbf{c})$, on pairs $(\mathbf{x}_i, \mathbf{c}_i)$; loss $\mathrm{BCE}(\mathbf{x}, \sigma(G_\theta(\mathbf{z},\mathbf{c}))) + \mathrm{KL}$ as before
- $\mathbf{c}$ already tells the decoder *which* digit; the KL term charges every nat stored in $\mathbf{z}$, so $\mathbf{z}$ keeps only what $\mathbf{c}$ does not say: the **style** (slant, width, size)
- $\mathbf{c}$ can be any vector: a one-hot class, a temperature T, a point (g, T) of a phase diagram; the prior may also depend on it, $p_\psi(\mathbf{z}|\mathbf{c})$ (Section 6)

Generating at a Chosen Condition



Fix $\mathbf{c}$, draw $\mathbf{z} \sim p(\mathbf{z}|\mathbf{c}) = \mathcal{N}(0, I)$, decode: samples of $p_\theta(\mathbf{x}|\mathbf{c})$, all the ways of writing this digit. Same $\mathbf{z}$, different $\mathbf{c}$: same style, different digit.

Digits on Demand

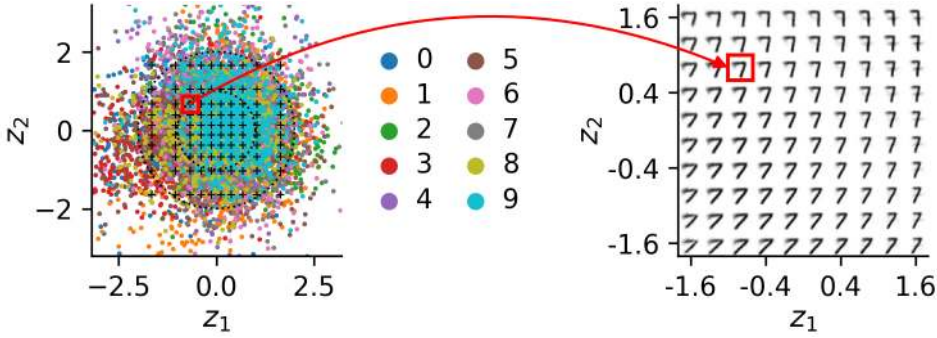


$\sigma(G_\theta(\mathbf{z}, \mathbf{c}))$; row: $\mathbf{c}$ = digit 0 to 9; column: one $\mathbf{z} \sim \mathcal{N}(0, I)$, the same in every row. Along a column the digit changes and the style stays.



Style transfer: a real test digit $\mathbf{x}$ (left), $\mathbf{z} = \boldsymbol{\mu}_\phi(\mathbf{x}, \mathbf{c}_y)$, decoded with $\mathbf{c} = 0, \ldots, 9$.

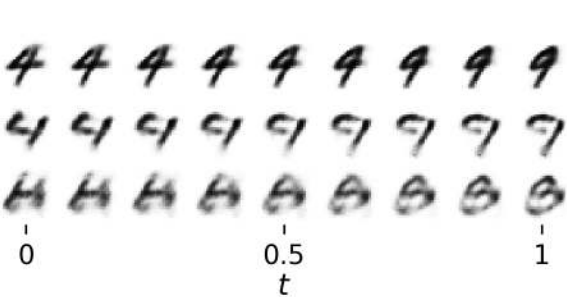
The Latent Plane of a cVAE Holds Only the Style

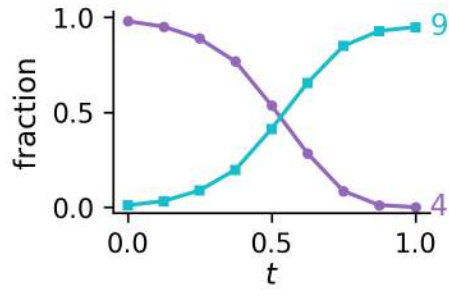


cVAE, $d = 2$, layout and quantile grid as before. Left: $\boldsymbol{\mu}_\phi(\mathbf{x}, \mathbf{c})$ of the test digits, coloured by label; right: **cell = decoder mean with $\mathbf{c} = 7$** at that grid point.

- all digits share one disc: digit predicted from the nearest class mean in the latent plane, VAE 0.73, cVAE 0.18 (chance 0.1)
- every cell is a 7; moving across the grid changes only how it is written

A Condition Never Seen in Training Is Interpolated

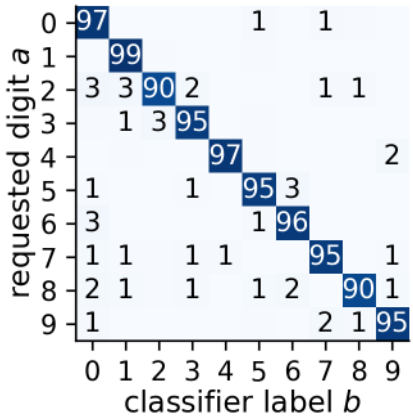


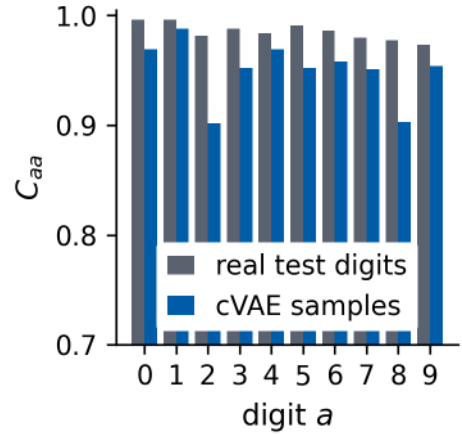


$\mathbf{c}(t) = (1-t)\,\mathbf{e}_4 + t\,\mathbf{e}_9$ ($\mathbf{e}_k$: one-hot vector of digit k); left: three fixed $\mathbf{z}$, right: fraction of 1000 samples, $\mathbf{z} \sim \mathcal{N}(0, I)$, that the classifier assigns to 4 and to 9.

- only $t = 0, 1$ were seen; in between the decoder interpolates, the style of each row is kept
- for digits there is no "6.5"; for a temperature or a point (g, T) intermediate values exist, and interpolated samples can be checked against Monte Carlo (Section 6)

The Requested Digit Is Generated





1000 samples per requested digit, $\mathbf{z} \sim \mathcal{N}(0, I)$;
$C_{ab} = \#\{\text{samples generated with } \mathbf{c} = a \text{ assigned to } b\}/1000$, in %. Right: C_{aa} next to the classifier on real test digits. Mean C_{aa}: 0.950; largest confusion $2 \to 0$: 3%.

From Digits to Spins

The protocol of this part, ready for physics:

1. a generator conditioned on what we want: $\mathbf{c}$ = the digit $\longrightarrow$ $\mathbf{c} = T$ for the 2D Ising model, $\mathbf{c} = (g, T)$ for the J_1–J_2 model
2. referees trained on **real** data, checked on real test data first $\longrightarrow$ networks trained on Monte Carlo configurations
3. the same tests on synthetic samples

Physics has a stricter referee

the true distribution is known, $p(\mathbf{s} \mid T) = e^{-H(\mathbf{s})/T}/Z$: energy, magnetisation, fluctuations and correlations of generated configurations can be compared with Monte Carlo at every T

The test

cVAE samples against the Monte Carlo samples they were trained on, observable by observable, including near the phase transition

6

Generated Spins and the Hamiltonian

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The Hamiltonian Fixes the Distribution

spins $s_i = \pm 1$, $L \times L$ lattice, periodic boundaries, $N = L^2$, $k_B = J = J_1 = 1$, $\beta = 1/T$:

$$H_{\text{Ising}} = -\sum_{\langle ij \rangle} s_i s_j, \quad H_{J_1 J_2} = -\sum_{\langle ij \rangle} s_i s_j + g \sum_{\langle\langle ij \rangle\rangle} s_i s_j, \quad p(\mathbf{s} \mid T) = \frac{e^{-\beta H(\mathbf{s})}}{Z}$$



J_1, $J_2 = g$

ground states ($T = 0$), $2N$ nearest and $2N$ diagonal bonds: ferromagnet $e = -2 + 2g$; stripes (sketch: nearest bonds cancel, diagonals antiparallel) $e = -2g$; equal at $g = \frac{1}{2}$: ferromagnet for $g < \frac{1}{2}$, stripes for $g > \frac{1}{2}$

2D Ising ($g = 0$): $T_c = 2/\ln(1 + \sqrt{2}) = 2.269$ (L. Onsager, Phys. Rev. **65**, 117 (1944); C. N. Yang, Phys. Rev. **85**, 808 (1952)). A generator of spin configurations has to reproduce $p(\mathbf{s} \mid T)$, not only its look.

Observables Are Averages over the Boltzmann Distribution

- every observable is an average over p: $\langle O \rangle = \frac{1}{Z} \sum_{\mathbf{s}} O(\mathbf{s})\, e^{-\beta H(\mathbf{s})}$
- $\ln Z$ generates the thermodynamics: $\langle e \rangle = -\frac{1}{N} \frac{\partial \ln Z}{\partial \beta}$, $c = \frac{\beta^2}{N} \frac{\partial^2 \ln Z}{\partial \beta^2} = N\beta^2 \left(\langle e^2 \rangle - \langle e \rangle^2 \right)$
- the sum has 2^N terms ($2^{900} \approx 8.5 \times 10^{270}$): Monte Carlo draws $\mathbf{s}^{(1)}, \ldots, \mathbf{s}^{(n)} \sim p$, $\langle O \rangle \approx \overline{O} = \frac{1}{n} \sum_k O(\mathbf{s}^{(k)})$
- a generator is judged by the **same** sample means over its own configurations

T	1.50	2.00	2.26	2.50	3.00	3.50
$\overline{e}$	-1.95	-1.74	-1.43	-1.10	-0.82	-0.66
$\overline{\lvert m \rvert}$	0.99	0.91	0.64	0.19	0.09	0.06
c	0.20	0.51	1.94	0.76	0.34	0.23

Ising, $L = 30$, $n = 64$ Swendsen–Wang snapshots per T: $\overline{|m|}$ drops and c peaks near $T_c = 2.269$

Observables: Definition and Estimator from Samples

$e_k = H(\mathbf{s}^{(k)})/N$, $m_k = \frac{1}{N}\sum_i s_i^{(k)}$: energy and magnetisation per spin of sample k; $\overline{x} = \frac{1}{n}\sum_k x_k$; $\langle m\rangle = 0$ on a finite lattice (symmetry $\mathbf{s} \to -\mathbf{s}$), hence $|m|$

	definition through Z	**estimator from n samples**	**measures**
energy	$\langle e\rangle = -\frac{1}{N}\partial_\beta \ln Z$	$\overline{e}$	bonds
magnetisation	$\langle\lvert m\rvert\rangle = \frac{1}{Z}\sum_{\mathbf{s}} \lvert m(\mathbf{s})\rvert\, e^{-\beta H}$	$\overline{\lvert m\rvert}$	ferromagnetism
stripe order	$\langle m_{\mathrm{str}}\rangle$, $m_{\mathrm{str}} = \left(m^2_{(\pi,0)} + m^2_{(0,\pi)}\right)^{1/2}$	$\overline{m_{\mathrm{str}}}$	stripes
specific heat	$c = N\beta^2\left(\langle e^2\rangle - \langle e\rangle^2\right)$	$N\beta^2\left(\overline{e^2} - \overline{e}^2\right)$	e fluct.
susceptibility	$\chi = N\beta\left(\langle m^2\rangle - \langle\lvert m\rvert\rangle^2\right)$	$N\beta\left(\overline{m^2} - \overline{\lvert m\rvert}^2\right)$	m fluct.
Binder cumulant	$U_4 = 1 - \langle m^4\rangle / 3\langle m^2\rangle^2$	$1 - \overline{m^4}/3\,\overline{m^2}^2$	$P(m)$ shape
correlation	$G(r) = \frac{1}{2N}\sum_{\mathbf{x},\mu}\langle s_{\mathbf{x}} s_{\mathbf{x}+r\hat{\mathbf{e}}_\mu}\rangle$	same with $\overline{\cdot}$	range
distributions	$P(e') = \frac{1}{Z}\sum_{\mathbf{s}}\delta\left(e' - H/N\right)e^{-\beta H}$	histogram of e_k (and m_k)	all moments

$m_{(\pi,0)} = \frac{1}{N}\sum_{\mathbf{x}}(-1)^x s_{\mathbf{x}}$, $m_{(0,\pi)}$ with $(-1)^y$; $\hat{\mathbf{e}}_\mu$: lattice unit vectors; variances divide by n

The cVAE for Spins: a Product of Bernoulli Factors

condition $c = T$ (Ising) or $c = (g, T)$ (J_1–J_2); bits $b_i = (s_i + 1)/2$; one logit $\ell_i(\mathbf{z}, c)$ per site, $\sigma(\ell_i)$ = probability that spin i points up

$$\log p_\theta(\mathbf{s} \mid c) \;\geq\; \mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{s},c)}\Big[\sum_{i=1}^{N} b_i \log \sigma(\ell_i) + (1 - b_i) \log\Big(1 - \sigma(\ell_i)\Big)\Big]$$
$$- \,\mathrm{KL}\Big(q_\phi(\mathbf{z} \mid \mathbf{s}, c)\,\Big\|\, p(\mathbf{z} \mid c)\Big) \;=\; \mathrm{ELBO}$$

the first term is $-$BCE of the spins ($N \ln 2 = 624$ nats for coin flips, $N = 900$); prior $p(\mathbf{z}|c) = \mathcal{N}(0, I)$ as in the MNIST cVAE, or learned (below)

trained Ising models ($L = 30$, $d = 16$): $-$ELBO $\approx$ 363–391 nats per configuration (training, held-out), against $N \ln 2 = 624$ for independent coin flips

D. P. Kingma, M. Welling, arXiv:1312.6114; K. Sohn, H. Lee, X. Yan, NeurIPS 2015, NeurIPS proceedings

The cVAE and Monte Carlo Sample Spins Differently

Generating at temperature T

$\mathbf{z} \sim p(\mathbf{z} \mid T)$, logits $\ell_i(\mathbf{z}, T)$, then every spin *independently*: $s_i = +1$ with probability $\sigma(\ell_i)$ (Bernoulli), or $\hat{s}_i = \operatorname{sign} \ell_i$ (most likely). Given $\mathbf{z}$ a mean-field-like product, $\langle s_i \rangle_{\mathbf{z}} = \tanh(\ell_i/2)$,

$$\operatorname{Cov}(s_i, s_j) = \operatorname{Cov}_{\mathbf{z}}\left(\tanh \tfrac{\ell_i}{2}, \tanh \tfrac{\ell_j}{2}\right), \; i \neq j$$

all correlations pass through the $d = 16$ latent variables

R. H. Swendsen, J.-S. Wang, PRL **58**, 86 (1987)

Monte Carlo

a Markov chain of updates built from H (Metropolis single-spin flips, accepted with $\min(1, e^{-\beta \Delta H})$; Ising data here: Swendsen–Wang cluster flips); stationary distribution exactly $e^{-\beta H}/Z$, with every neighbour correlation H implies; costs equilibration, successive snapshots correlated. $H = -\sum s_i s_j$: independent draws break bonds at random $\Rightarrow$ e, c, $G(1)$ are the sharp tests

Every Test Has a Real-vs-Real Baseline

2D Ising, $L = 30$, 37 temperatures $T \in [1.5, 3.5]$, 256 Swendsen–Wang snapshots per T, split into **training** (128), **held-out** A (64), **held-out** B (64); synthetic: 64 snapshots per T



Noise units

$R_O = \mathrm{mean}_T|O_{\mathrm{syn}} - O_A| \,/\, \mathrm{mean}_T|O_B - O_A|$; $R \approx 1$: as close to A as an independent Monte Carlo set. Denominators: e 0.011, $|m|$ 0.018, c 0.17, χ 1.6

every test runs first on held-out Monte Carlo data, then identically on the synthetic data at the same T

Three Kinds of Tests

1. Networks trained on Monte Carlo data

CNN regressor $\hat{T}(\mathbf{s})$, loss $(\hat{T} - T)^2$; CNN phase classifier trained at $T \leq 1.8$ and $T \geq 2.8$, confidence $\max(p_0, p_1)$

2. The Hamiltonian

$\langle e \rangle$, $\langle |m| \rangle$, c, χ, U_4, $G(r)$ and the distributions $P(e)$, $P(m)$, compared by $W_1 = \frac{1}{n} \sum_k |x_{(k)} - y_{(k)}|$ ($x_{(k)}$: k-th smallest value)

C2ST: D. Lopez-Paz, M. Oquab, arXiv:1610.06545

3. The generator itself

$-$ELBO on training and held-out data; active latent units ($\mathrm{KL}_j > 0.01$); classifier two-sample test (C2ST): accuracy of a CNN separating real from synthetic, 0.5 = indistinguishable; memorisation: $d_{\mathrm{NN}} = \min_{\mathbf{t} \in \mathrm{train}} (1 - |\mathbf{s} \cdot \mathbf{t}|/N)/2$

Three cVAEs, Each Adding One Piece of Physics

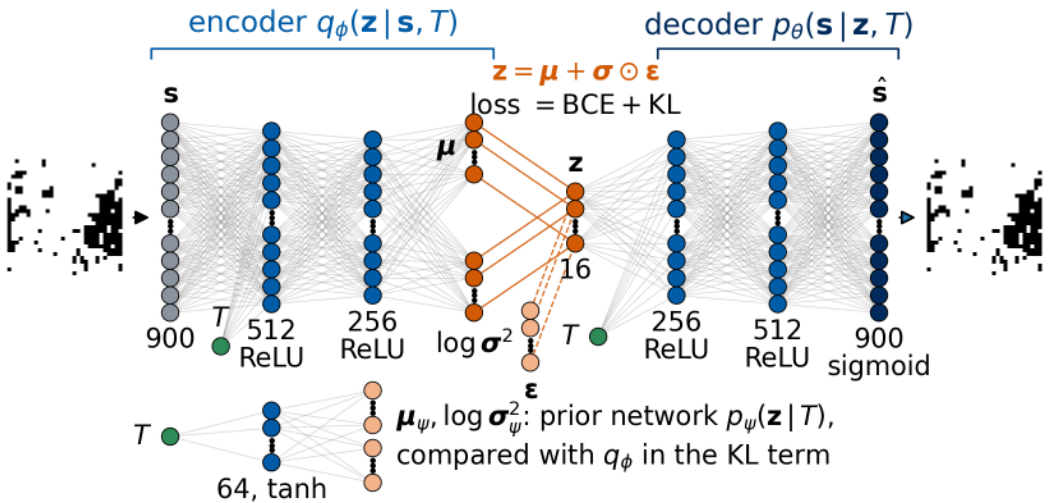
dense encoder $N{+}1 \to 512 \to 256 \to (\boldsymbol{\mu}, \log \boldsymbol{\sigma}^2)$, latent dimension $d = 16$; dense decoder $d{+}1 \to 256 \to 512 \to N$ logits; 40 epochs, Adam

- **cVAE**: prior $\mathcal{N}(0, I)$, raw data (the MNIST model)
- **cVAE+Z2**: the symmetry $H(-\mathbf{s}) = H(\mathbf{s})$ built in: train on the $m \geq 0$ representative of every snapshot, flip each generated snapshot with probability $\frac{1}{2}$
- **cVAE+Z2+p(z|T)**: conditional prior $p_\psi(\mathbf{z} \mid T) = \mathcal{N}(\boldsymbol{\mu}_\psi(T), \operatorname{diag} \boldsymbol{\sigma}^2_\psi(T))$ from a small network $T \to 64 \to (\boldsymbol{\mu}_\psi, \log \boldsymbol{\sigma}^2_\psi)$: a snapshot reveals its T, so the encoded snapshots of different T occupy different regions of $\mathbf{z}$-space; $\mathbf{z} \sim \mathcal{N}(0, I)$ draws from all of them, $\mathbf{z} \sim p_\psi(\mathbf{z}|T)$ from the region of T

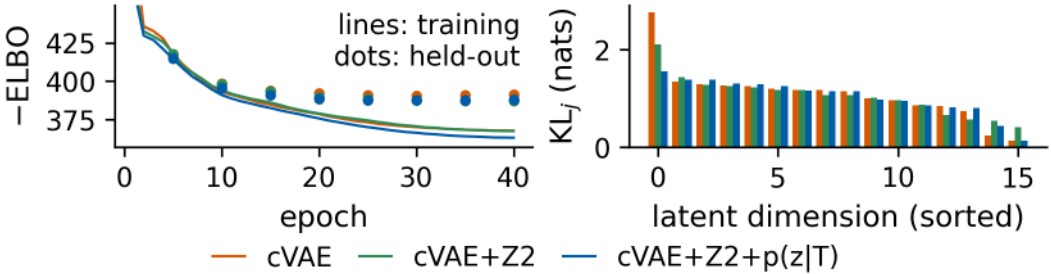
$$\mathrm{KL}(q \,\|\, p_\psi) = \frac{1}{2} \sum_{j=1}^{d} \Big[\log \frac{\sigma^2_{\psi,j}}{\sigma^2_j} + \frac{\sigma^2_j + (\mu_j - \mu_{\psi,j})^2}{\sigma^2_{\psi,j}} - 1 \Big]$$

prior network: K. Sohn, H. Lee, X. Yan, NeurIPS 2015, NeurIPS proceedings

The Spin cVAE with a Prior Network



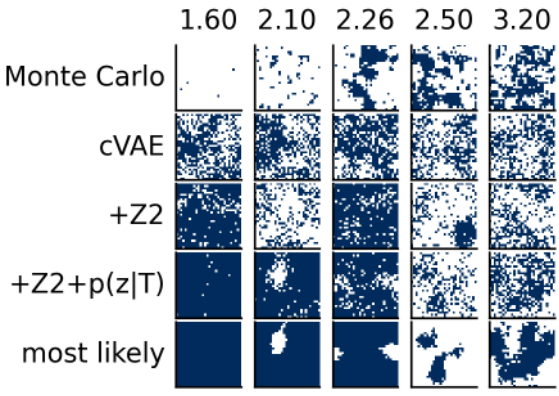
Training Health Does Not Rank the Three Models



left: −ELBO per configuration (nats; uniform spins $N \ln 2 = 624$); right: held-out KL_j per latent dimension

	cVAE	cVAE+Z2	cVAE+Z2+p(z|T)
−ELBO training / held-out	368 / 391	368 / 387	363 / 388
active units ($\mathrm{KL}_j > 0.01$) of 16	16	16	16
spins reproduced from $\mathbf{z} = \boldsymbol{\mu}$	80 %	81 %	81 %

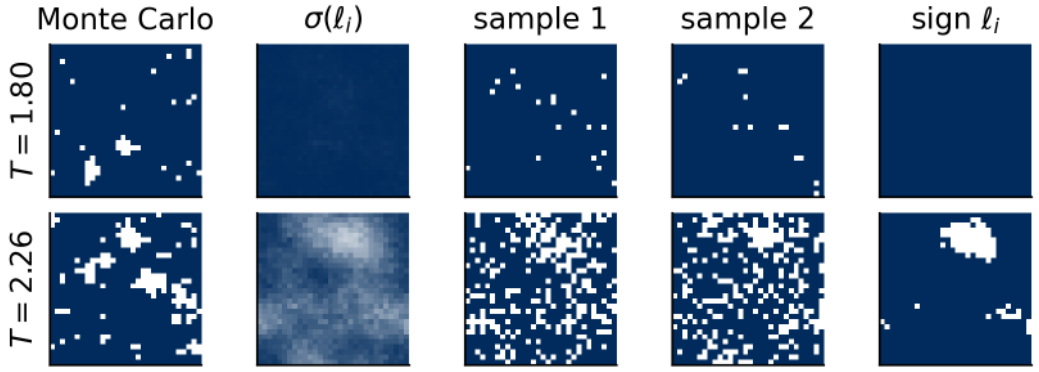
Monte Carlo and Synthetic Ising Configurations



rows: Monte Carlo; cVAE; cVAE+Z2; cVAE+Z2+p(z|T) with Bernoulli sampling; the same $\mathbf{z}$ with $\hat{s}_i = \operatorname{sign} \ell_i$

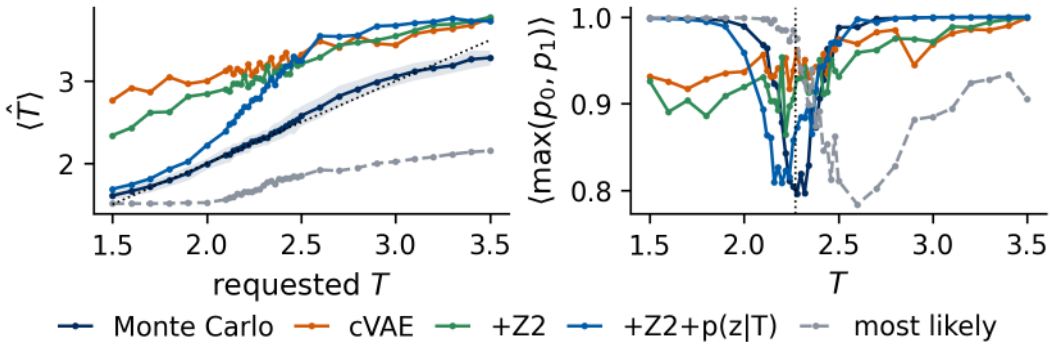
- cVAE and +Z2: the requested T is barely visible
- +p(z|T): ordered below T_c, disordered above; salt-and-pepper noise inside domains
- most likely configuration: no noise, but large smooth domains even at $T = 3.2$

Generated Spins Are Independent Coin Flips Given z



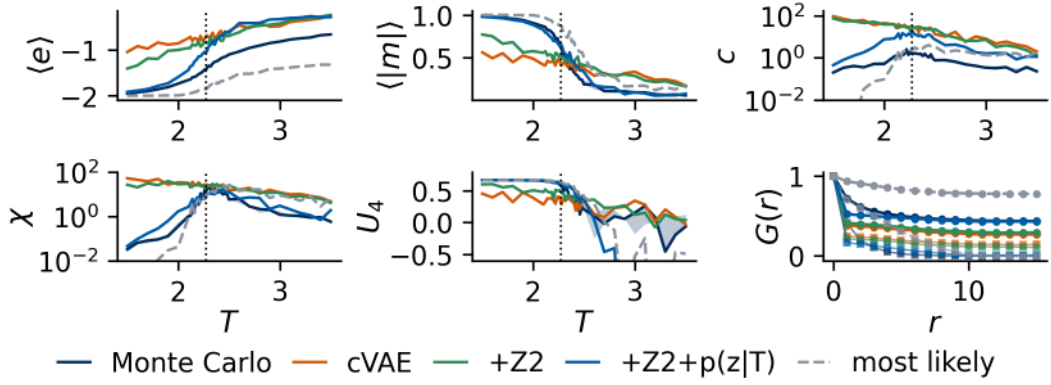
One $\mathbf{z} \sim p_\psi(\mathbf{z}|T)$ per row: probabilities $\sigma(\ell_i)$ (white 0, navy 1), two Bernoulli samples, $\operatorname{sign} \ell_i$. Nearest-neighbour correlation $G(1)$ at $T = 1.80$ / 2.26: samples 0.894 / 0.520, independent-spin prediction $\overline{\tanh \frac{\ell_i}{2} \tanh \frac{\ell_j}{2}}$ 0.895 / 0.521, Monte Carlo 0.930 / 0.717: **fewer satisfied bonds than $e^{-\beta H}/Z$**; the magnetisation averages the coin flips out.

Networks Trained on Monte Carlo Data Judge the Samples



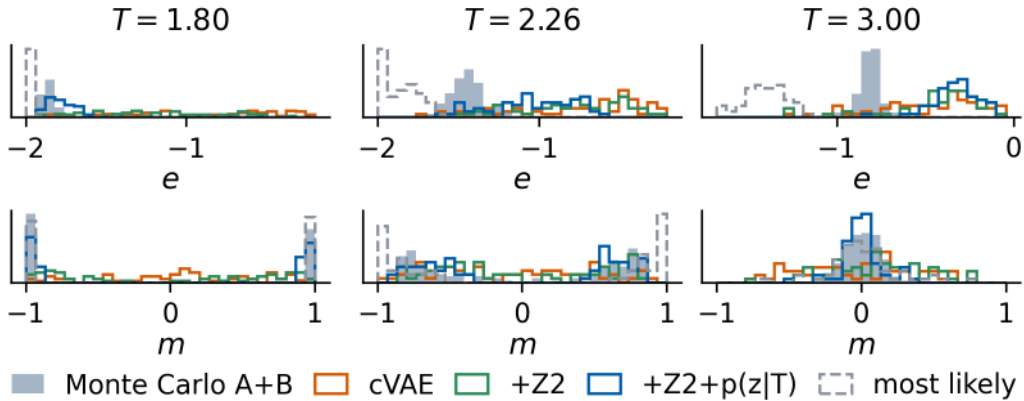
left: $\langle \hat{T} \rangle$ vs requested T (band: $\pm$ std on A); right: $\langle \max(p_0, p_1) \rangle_T$. Root-mean-square error (RMSE) of $\hat{T}$: Monte Carlo A 0.12, B 0.12; cVAE 0.98; +Z2 0.84; +p(z|T) 0.62; most likely 0.71. Below T_c only +p(z|T) is read at the requested temperature; above T_c every Bernoulli model looks too hot.

The Order Parameter Is Reproduced, the Energy Is Not



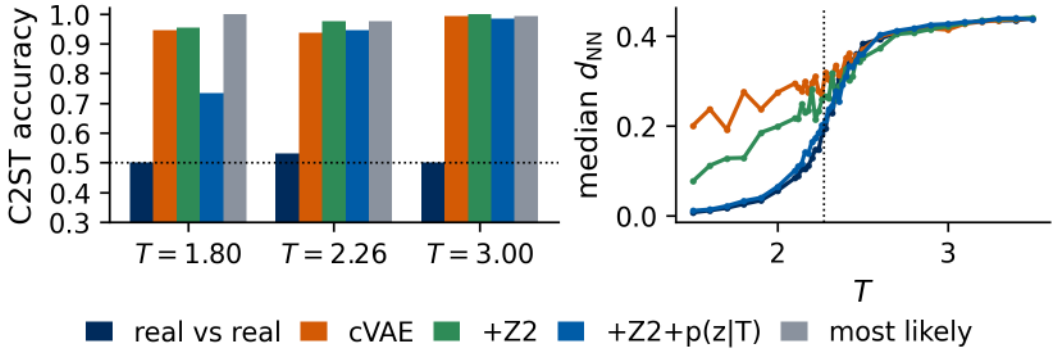
$\langle e \rangle$, $\langle |m| \rangle$, c, χ, U_4 vs T; $G(r)$ at $T = 2.26$ (circles) and $T = 3.0$ (squares); band: Monte Carlo A vs B. Noise ratios of +p(z|T): $R_{|m|} = 2.8$, $R_\chi = 2.0$, $R_{W_1(m)} = 1.3$; $R_e = 34.8$, $R_c = 39.0$, $R_{W_1(e)} = 21.6$. At $T = 2.26$: $\langle e \rangle = -1.43$ (Monte Carlo), -1.04 (Bernoulli), -1.86 (most likely); $G(1) = -\langle e \rangle / 2$: the energy is the nearest-neighbour correlation

Distributions of Energy and Magnetisation



Histograms $P(e)$, $P(m)$ at $T = 1.8, 2.26, 3.0$; Wasserstein distances to Monte Carlo A in noise units: $R_{W_1(e)} = 21.6$, $R_{W_1(m)} = 1.3$ (+p(z|T)). Fraction of snapshots below T_c with $|m| < \langle|m|\rangle_{\mathrm{MC}}/2$: Monte Carlo 1 %, cVAE 45 %, +Z2 27 %, +p(z|T) 1 %. $P(m)$ agrees within 1.3 noise units, $P(e)$ is shifted and too broad.

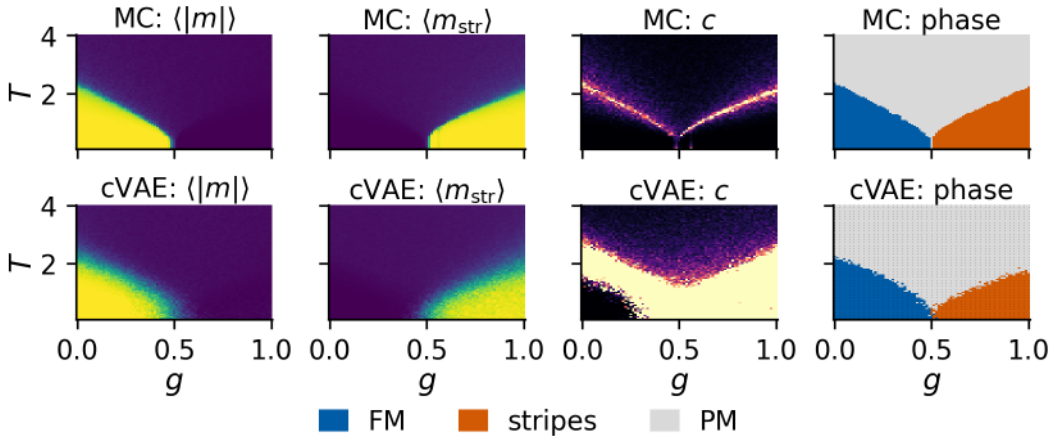
A Classifier Separates Real from Synthetic; Nothing Is Copied



C2ST held-out accuracy at $T = 1.8/2.26/3.0$: real vs real 0.50 / 0.53 / 0.50; +p(z|T) 0.73 / 0.95 / 0.98; most likely 1.00 / 0.98 / 0.99

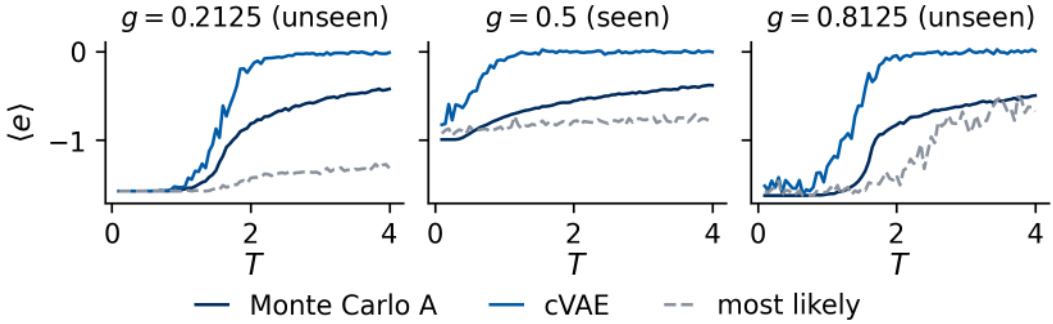
median d_{NN} to the training set at the same T: Monte Carlo 0.03 / 0.18 / 0.43; +p(z|T) 0.03 / 0.20 / 0.43: new configurations, no memorisation

J_1–J_2: the Phase Diagram Is Interpolated



Trained at every second g, T (dots); colour 0 to 1 (c: to 2.2); phase: CNN vote. Same phase as MC A: MC B 99.7 %, cVAE 95.3 % (seen) / 95.4 % (unseen), most likely 35.4 %.

J_1–J_2: Order Parameters Close, Energy Far Off



R (noise units)	$\langle \lvert m \rvert \rangle$	$\langle m_{\mathrm{str}} \rangle$	$\langle e \rangle$	c
seen (g, T)	4.4	6.4	64.0	2279
unseen (g, T)	4.8	5.9	63.9	1517

−ELBO held-out: 476 (seen), 482 (unseen); stripes alternating along x: 0.50 (Monte Carlo), 0.48 (cVAE). Paramagnet: synthetic $\langle e \rangle \approx 0$, the value for independent spins

The Generator Reproduces the Phase

Reproduced within a few noise units

- Ising $\langle|m|\rangle$, χ, $P(m)$: $R = 2.8, 2.0, 1.3$; J_1–J_2 $\langle|m|\rangle$, $\langle m_{\mathrm{str}}\rangle$: $R \approx 4.8, 5.9$ (unseen)
- the temperature seen by a regressor below T_c, the phase seen by classifiers trained on real data
- the J_1–J_2 phase diagram at unseen (g, T), both stripe orientations; new configurations, not copies

needed: $\mathbb{Z}_2$ symmetry of H by construction, and a conditional prior $p(\mathbf{z}|c)$. Ising; mixed: fraction of snapshots at $T < T_c$ with $|m| < \langle|m|\rangle_{\mathrm{MC}}/2$; Monte Carlo: RMSE 0.12, mixed 1 %

	$R_{\lvert m\rvert}$	R_χ	$R_{W_1(m)}$	RMSE $\hat{T}$	mixed
cVAE	13.3	11.8	3.3	0.98	45 %
cVAE+Z2	10.8	9.7	2.8	0.84	27 %
+p(z|T)	2.8	2.0	1.3	0.62	1 %

The Generator Does Not Reproduce the Energy

Not reproduced

- $\langle e \rangle = H/N$: $R = 34.8$ (Ising), 63.9 (J_1–J_2)
- c: 2 to 15 times the Monte Carlo value (Ising, six tabulated T); $P(e)$ too broad
- a C2ST separates real from synthetic at the three tested T (accuracy 0.73–0.98)

given $\mathbf{z}$ the decoder is a product over sites, a mean-field-like distribution: nearest-neighbour correlations beyond what 16 latent variables carry are lost. Bernoulli sampling breaks bonds, $\operatorname{sign} \ell_i$ over-smooths.

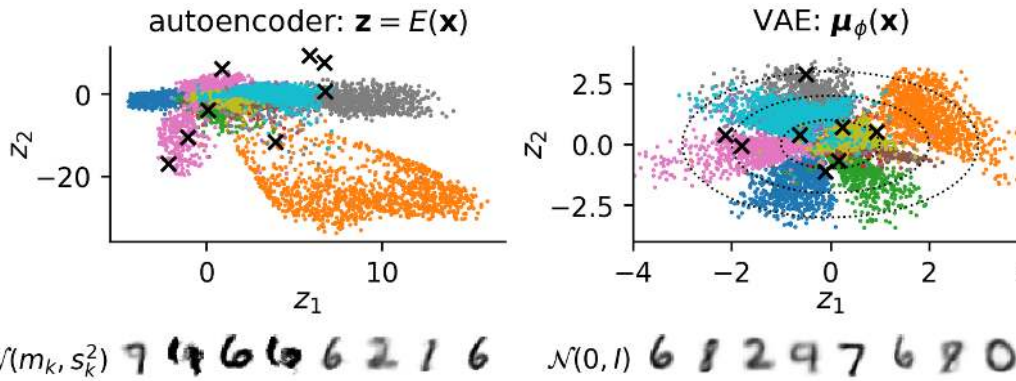
Looking right is not obeying H: synthetic configurations answer questions about the phase; they are not samples of $e^{-H/T}/Z$ and do not replace Monte Carlo for thermodynamics.

7

Disentangled Representations: the β-VAE

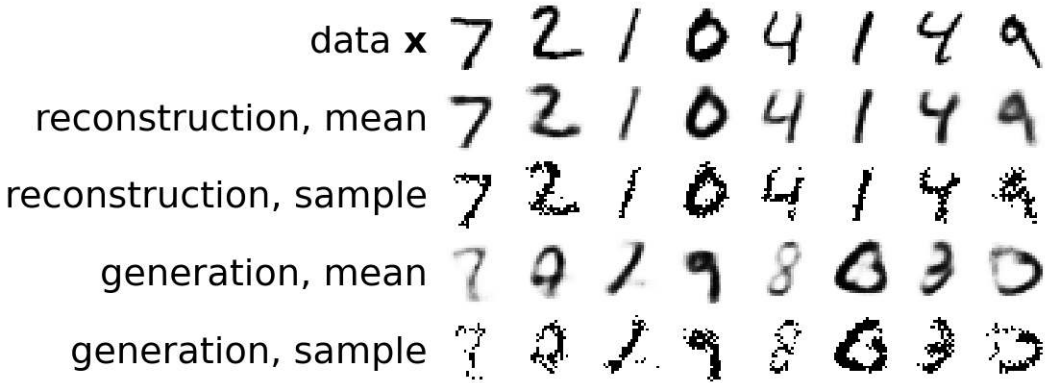
JAGIELLONIAN UNIVERSITY
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An Autoencoder Cannot Be Sampled, a VAE Can



$d = 2$, 10 000 test digits (colour = label, for plotting only). Left: AE codes $\mathbf{z} = E(\mathbf{x})$; right: VAE means $\boldsymbol{\mu}_\phi(\mathbf{x})$, dotted $|\mathbf{z}| = 1, 2, 3$. Crosses: random $\mathbf{z}$, decoded below. AE: standard deviations 3.84, 7.61, islands with empty space, 2 of 8 decodes are blobs. VAE: 1.06, 1.09, the disc is filled, all 8 decodes are digits.

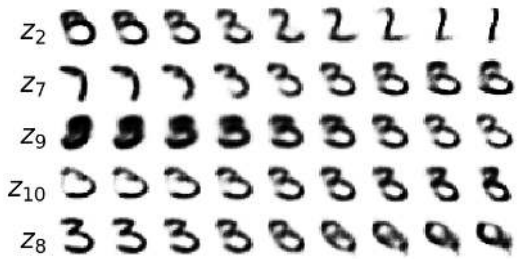
Sampling a Trained VAE: Reconstruction and Generation



reconstruction: $\mathbf{x} \to \mathbf{z} \sim q_\phi(\mathbf{z}|\mathbf{x}) \to$ decoder; **generation**: $\mathbf{z} \sim \mathcal{N}(0, I) \to \mathbf{x} \sim p_\theta(\mathbf{x}|\mathbf{z})$, a draw from $\int p(\mathbf{z})\, p_\theta(\mathbf{x}|\mathbf{z})\, d\mathbf{z}$

mean $\hat{x}_j = \sigma(G_\theta(\mathbf{z}))_j$: blurry (uncertain = grey); **sample**: pixel j black with probability $\hat{x}_j$, independently: noisy. Figures show the mean.

In a Plain VAE One Coordinate Changes Several Features



VAE, $d = 10$: $\boldsymbol{\mu}_\phi(\mathbf{x})$ of one test 3; row: one z_k from -3 to 3, others fixed

Latent traversal: sampling along one axis, $\mathbf{z} = (\mu_1, \ldots, \mu_{k-1}, t, \mu_{k+1}, \ldots)$, $t \in [-3, 3]$, decoded.

- z_2: $3 \to 2 \to 1$ with slant; z_7: $7 \to 3 \to 8$; only z_9 changes one property (thickness)
- tells: what the decoder does along this axis *near this code*
- does not tell: that the axis means the same everywhere, or that no other axis changes the same feature

The Goal: One Coordinate per Independent Factor

Generative factors: the few independent quantities that produced an image.
sprite: shape, scale, rotation, x, y · face: pose (looking left or right), glasses, smile, hair · digit: identity, slant, thickness · spins: magnetisation m, stripe orientation

Disentangled code $\mathbf{z} = (z_1, \ldots, z_d)$: each factor is carried by its own coordinate; moving one z_k changes one property of the image, and changing one factor changes one z_k.

- *control* (add glasses by moving one number), *interpretation*, *discovery* (an order parameter without labels)
- cVAE: the factor is *given* as a label $\mathbf{c}$; here the network must *discover* it

MNIST	watch the loss at work: rate, distortion, samples, coordinates switching off
dSprites	synthetic sprites with 5 known factors: *measure* disentangling
CelebA	CelebFaces Attributes: celebrity faces with 40 labels; traversals, arithmetic
spins	Ising, J_1–J_2: which coordinate is the magnetisation, the energy, the stripes?

Three Quantities Behind the Loss

prior $p(\mathbf{z}) = \mathcal{N}(0, I_d)$; encoder $q_\phi(\mathbf{z}|\mathbf{x}) = \mathcal{N}(\boldsymbol{\mu}_\phi(\mathbf{x}), \operatorname{diag} \boldsymbol{\sigma}^2_\phi(\mathbf{x}))$; decoder $p_\theta(\mathbf{x}|\mathbf{z})$: independent Bernoulli pixels with probabilities $\hat{\mathbf{x}} = \sigma(G_\theta(\mathbf{z}))$

- **BCE** (binary cross-entropy), the reconstruction error:

$$\mathrm{BCE}(\mathbf{x}, \hat{\mathbf{x}}) = -\sum_j \left[x_j \log \hat{x}_j + (1 - x_j) \log(1 - \hat{x}_j)\right] = -\log p_\theta(\mathbf{x}|\mathbf{z})$$

  nats needed to explain the pixels of $\mathbf{x}$; 0 for a perfect, confident prediction; $784 \log 2 = 543$ for $\hat{x}_j = \frac{1}{2}$

- **KL** (Kullback–Leibler divergence): $\mathrm{KL}(q\|p) = \mathbb{E}_q[\log q - \log p] \geq 0$, $= 0$ iff $q = p$; here: how far the cloud of $\mathbf{x}$ is from the prior, in nats, $\frac{1}{2}\sum_k(\mu_k^2 + \sigma_k^2 - \ln \sigma_k^2 - 1)$

- **ELBO** (evidence lower bound): $\mathbb{E}_q[\log p_\theta(\mathbf{x}|\mathbf{z})] - \mathrm{KL} \leq \log p_\theta(\mathbf{x})$, a computable lower bound on the log-likelihood (the "evidence"), which itself is an intractable integral over $\mathbf{z}$

The β-VAE Loss, Term by Term

$$\mathcal{L}_\beta(\mathbf{x}) = \underbrace{\mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x})}\left[\mathrm{BCE}(\mathbf{x}, \hat{\mathbf{x}})\right]}_{\text{reconstruction error}} + \beta \underbrace{\mathrm{KL}\left(q_\phi(\mathbf{z}|\mathbf{x}) \,\|\, p(\mathbf{z})\right)}_{\text{code vs. prior}}, \qquad \beta = 1: \;\; \mathcal{L}_1 = -\mathrm{ELBO}$$

- reconstruction: samples of the cloud of $\mathbf{x}$ decode back to $\mathbf{x}$; KL: each cloud pulled to $\mathcal{N}(0, 1)$
- $\beta > 1$: information costs more $\Rightarrow$ fewer used coordinates, wider clouds, blurrier images, stronger pull to the *factorised* prior $\prod_k \mathcal{N}(0, 1)$: towards independent coordinates
- $\beta < 1$: sharper, but the code drifts from the prior; $\beta \to 0$: an autoencoder
- $\beta =$ Lagrange multiplier: best reconstruction at $\mathrm{KL} < \varepsilon$ (not $1/T$)

I. Higgins et al., ICLR 2017, OpenReview

Rate and Distortion: β Is Minus the Slope

Averages over the data $p_{\mathrm{data}}(\mathbf{x})$, in nats per image:

$$D = \mathbb{E}_{p_{\mathrm{data}}}\mathbb{E}_{q_\phi}\big[-\log p_\theta(\mathbf{x}|\mathbf{z})\big], \qquad R = \mathbb{E}_{p_{\mathrm{data}}}\mathrm{KL}(q_\phi(\mathbf{z}|\mathbf{x}) \,\|\, p(\mathbf{z})), \qquad \mathcal{L}_\beta = D + \beta R$$

- **rate** R: the information the code carries about the image; **distortion** D: the reconstruction error
- every trained model is a point (R, D); the reachable points lie above a decreasing curve $D^*(R)$
- minimising $D + \beta R$ selects the point where $dD^*/dR = -\beta$: β is minus the slope
- large β: the flat, low-rate end; small β: the steep, high-rate end
- the ELBO $= -(D + R)$ alone does not fix how a model splits its loss between R and D

A. A. Alemi et al., *Fixing a Broken ELBO*, ICML 2018, PMLR 80, 159

β Is a Temperature in a Free Energy

Lab 024: $-\mathrm{ELBO}$ is a variational free energy $F[q] = \langle E \rangle_q - TS[q]$. For one image, with the entropy $S[q] = -\mathbb{E}_q \log q$ of the code:

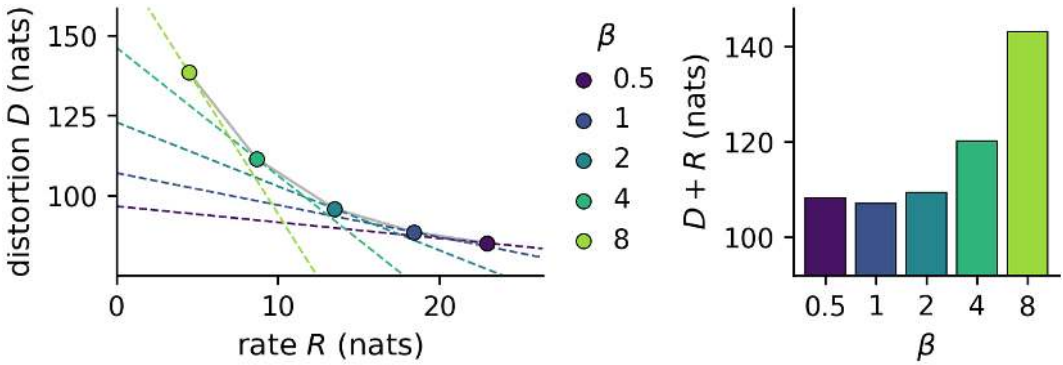
$$\mathcal{L}_\beta[q] = \langle E_\beta \rangle_q - \beta\, S[q], \qquad E_\beta(\mathbf{z}) = -\log p_\theta(\mathbf{x}|\mathbf{z}) - \beta \log p(\mathbf{z})$$

minimised by the Boltzmann distribution of E_β at temperature β, a *tempered posterior*:

$$q^*_\beta(\mathbf{z}) \propto e^{-E_\beta(\mathbf{z})/\beta} = p_\theta(\mathbf{x}|\mathbf{z})^{1/\beta}\, p(\mathbf{z})$$

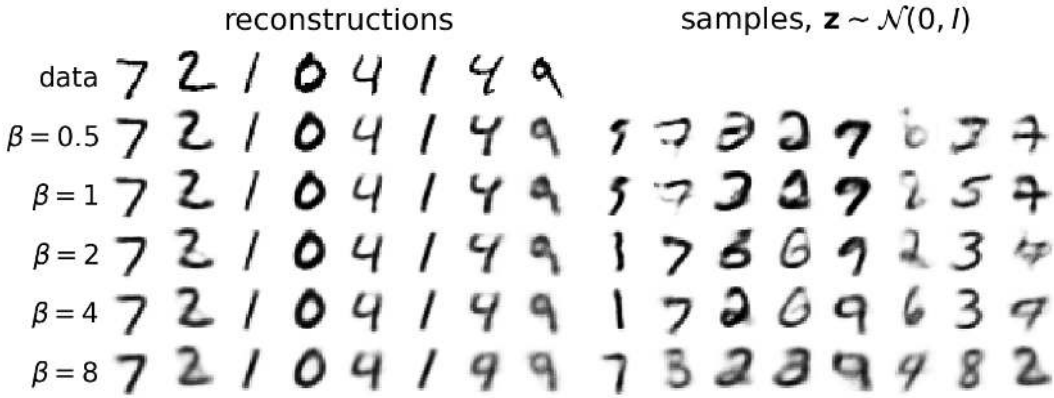
- β weights the entropy of the code, as T weights the entropy in thermodynamics; $\beta = 1$: the true posterior
- $\beta > 1$ (hot): broad, close to the prior, details of $\mathbf{x}$ forgotten; $\beta < 1$ (cold): concentrated
- checked on a Gaussian model (Lab 026): gradient descent reproduces q^*_β to all printed digits

Larger β Trades Distortion for Rate on MNIST



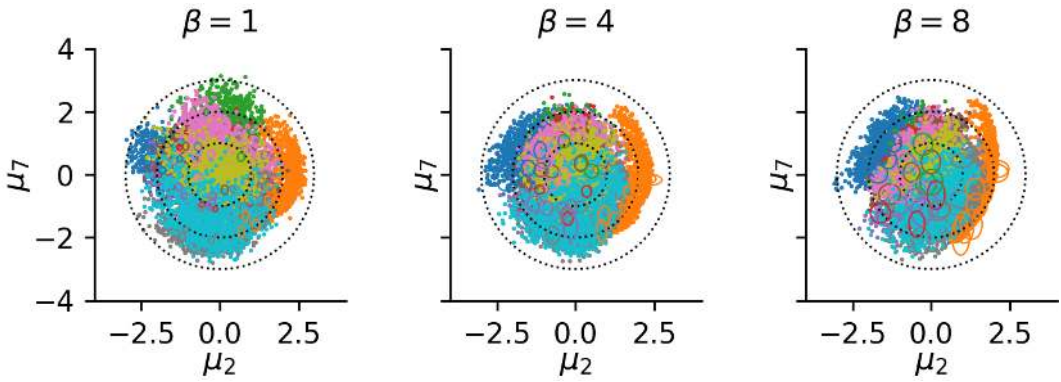
$d = 10$, $\beta \in \{0.5, 1, 2, 4, 8\}$, 10 000 test digits; $R = \langle \mathrm{KL}(q_\phi \| p) \rangle_{\mathbf{x}}$, $D = \langle \mathrm{BCE} \rangle_{\mathbf{x}}$; dashed: $D + \beta R = \text{const}$. R: 22.9 $\to$ 4.5, D: 85.2 $\to$ 138.5 nats. Best ELBO at $\beta = 1$ (107.1), but $\beta = 0.5, 2$ lie within 2.3 nats with 4.5 more or 4.9 fewer nats in the code.

Larger β Blurs Reconstructions and Samples



Left: test digits and $\sigma(G_\theta(\boldsymbol{\mu}_\phi(\mathbf{x})))$; right: $\sigma(G_\theta(\mathbf{z}))$, $\mathbf{z} \sim \mathcal{N}(0, I)$, the *same* 8 vectors $\mathbf{z}$ for every β. $\beta = 8$: detail lost (the 4 becomes a 9); the same $\mathbf{z}$ decodes to different digits in different models; $\beta = 0.5, 1$: some malformed samples.

Larger β Pushes the Clouds onto the Prior



Points: $\boldsymbol{\mu}_\phi(\mathbf{x})$ of the test digits in the two most active coordinates (colour = digit); ellipses: clouds $\boldsymbol{\mu} \pm \boldsymbol{\sigma}$ of 40 digits; dotted: $|\mathbf{z}| = 1, 2, 3$. $\beta = 1$: tiny, separated clouds; $\beta = 8$: wide, overlapping clouds, means squeezed (std 0.93 against 1.05, 1.09): detail is lost.

The Rate Splits into Three Parts

Which part of the rate pushes the coordinates of $\mathbf{z}$ to be independent? Look at all codes together: the **aggregate posterior** $q(\mathbf{z}) = \frac{1}{N}\sum_n q_\phi(\mathbf{z}|\mathbf{x}_n)$, with marginals $q(z_j)$; $n =$ index of the image.

$$R \;=\; \underbrace{I_q(n;\mathbf{z})}_{\text{index-code MI}} \;+\; \underbrace{\mathrm{KL}\Big(q(\mathbf{z}) \,\Big\|\, \textstyle\prod_j q(z_j)\Big)}_{\text{total correlation (TC)}} \;+\; \underbrace{\textstyle\sum_j \mathrm{KL}(q(z_j)\,\|\,p(z_j))}_{\text{dimension-wise KL}}$$

- **MI** (mutual information) $I(a;b) = \mathrm{KL}(P(a,b)\,\|\,P(a)P(b))$: what b tells about a, 0 iff independent; $I_q(n;\mathbf{z}) \le \log N$: how well the code identifies its image
- **TC** ≥ 0, $= 0$ iff $z_1, \ldots, z_d$ are independent over the data set; dimension-wise KL: each coordinate's distance from $\mathcal{N}(0,1)$

Derivation (multiply and divide inside the logarithm, twice): Lab 026, Section 2.4.

M. D. Hoffman, M. J. Johnson, *ELBO surgery*, NIPS 2016 Workshop AABI, PDF; R. T. Q. Chen et al., NeurIPS 2018, arXiv:1802.04942

Penalising the Total Correlation Alone: the β-TCVAE

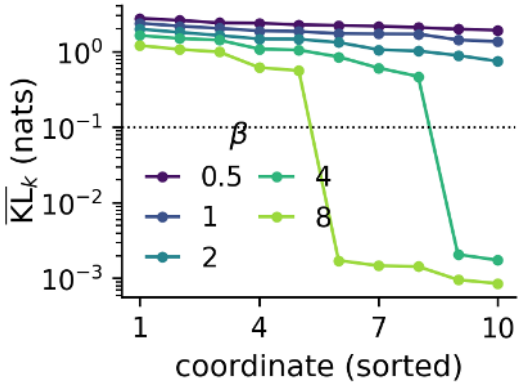
- the β-VAE multiplies all three parts by β; penalising I_q costs reconstruction quality
- if the factors that generate the data are independent, a code with one factor per coordinate has $\mathrm{TC} = 0$; a code mixing factors (e.g. rotated non-Gaussian coordinates) has $\mathrm{TC} > 0$
- β-TCVAE: weight β on the TC only,

$$\mathcal{L}_{\beta\text{-TC}} = D + I_q(n; \mathbf{z}) + \beta\,\mathrm{TC}(\mathbf{z}) + \textstyle\sum_j \mathrm{KL}(q(z_j) \,\|\, p(z_j))$$

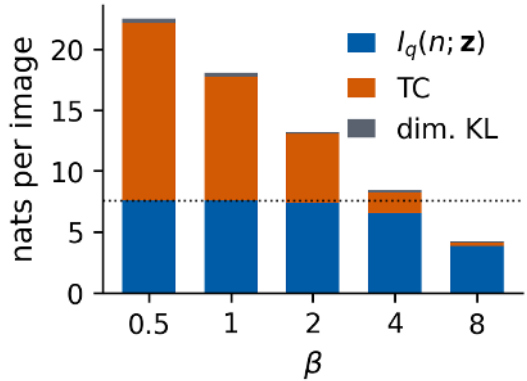
- $\log q(\mathbf{z})$ needs all N clouds; in training it is estimated on the minibatch, $\log q(\mathbf{z}_i) \approx \log \sum_{j \le B} q(\mathbf{z}_i|\mathbf{x}_j) - \log(NB)$
- FactorVAE: the same TC penalty, estimated with an auxiliary classifier

R. T. Q. Chen, X. Li, R. Grosse, D. Duvenaud, *Isolating Sources of Disentanglement in Variational Autoencoders*, NeurIPS 2018, arXiv:1802.04942; H. Kim, A. Mnih, *Disentangling by Factorising*, ICML 2018, PMLR 80, 2649

Larger β Switches Coordinates Off and Removes Total Correlation



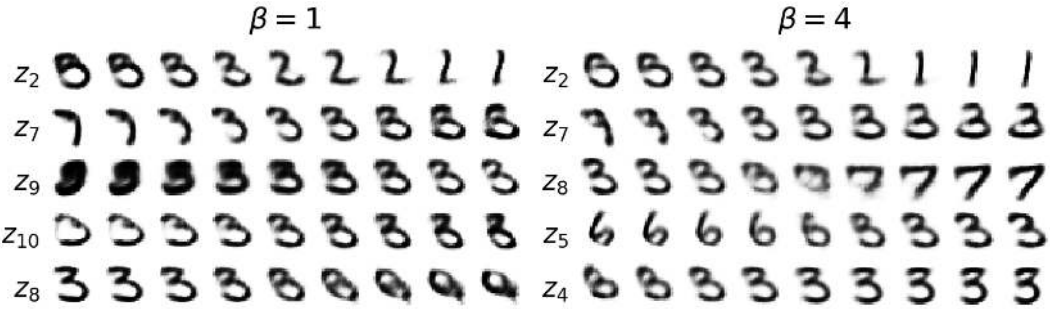
$\overline{\mathrm{KL}_k}$ sorted; active above 0.1 nats



parts of R on $M = 2000$ digits; dotted: $\log M$

Active: 10, 10, 10, 8, 5 of 10; TC: 14.50 $\rightarrow$ 0.26 nats; $I_q \approx \log M$ up to $\beta = 2$; parts add up to the closed-form R within 0.01 nats.

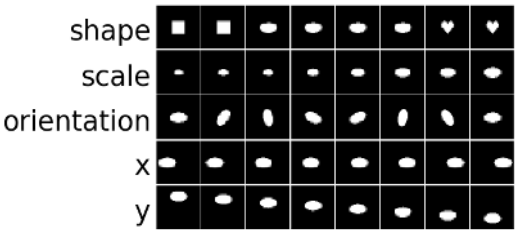
On MNIST, Traversals Show Style, Not Digit Identity



Same test 3; row: one active z_k from -3 to 3, others fixed.

- both models change the identity along every axis ($\beta = 4$: z_2: $6 \to 3 \to 1$, z_8: $3 \to 7$); $\beta = 4$ has fewer, somewhat cleaner axes (z_4: width, thickness)
- identity is *discrete*, $\mathbf{z}$ continuous: digits morph (cVAE: identity as a label); no true factors listed: disentangling cannot be *measured* on MNIST

dSprites: Known Factors, and the Mutual Information Gap



dSprites ("disentanglement testing sprites"): all $3 \cdot 6 \cdot 40 \cdot 32 \cdot 32 = 737\,280$ factor combinations; row: one factor varied; we use 75 000 images at 32×32

How much does each z_j know about each known factor v_k?

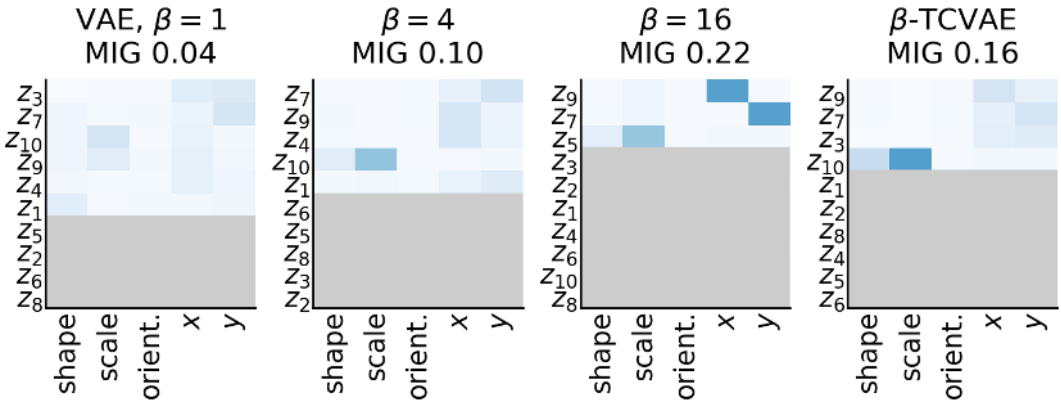
$$\mathrm{MIG} = \frac{1}{K} \sum_{k=1}^{K} \frac{I(z_{j_1}; v_k) - I(z_{j_2}; v_k)}{H(v_k)}$$

MIG = mutual information gap; j_1, j_2: coordinates with the largest and second-largest MI with v_k; $H = -\sum P \log P$: entropy; $z_j = \mu_j(\mathbf{x})$ in 20 bins; $K = 5$. 1: one coordinate per factor; 0: shared or absent.

Checks: true factors 0.90; x, y rotated by 45° 0.56; random 0.00.

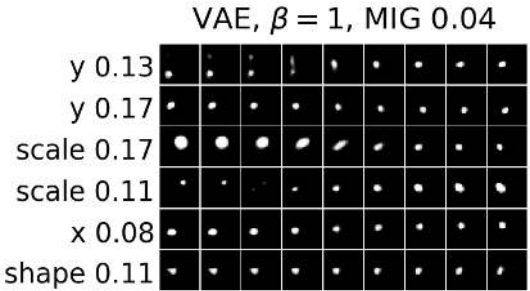
L. Matthey, I. Higgins, D. Hassabis, A. Lerchner, dSprites (2017), github.com/deepmind/dsprites-dataset; MIG: Chen et al. 2018

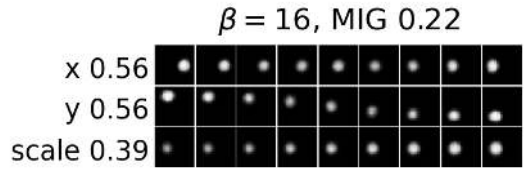
On dSprites, Larger β Raises the MIG from 0.04 to 0.22



Colour: $I(\mu_j; v_k)/H(v_k)$ from 0 to 1; rows: coordinates by $\overline{\mathrm{KL}}_j$, grey = inactive. $\beta = 16$: x, y, scale in separate coordinates; β-TCVAE: scale isolated (gap 0.51) at $D = 36$ not 64 nats.

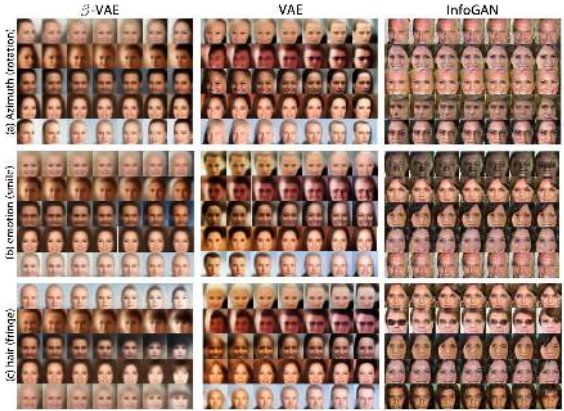
With $\beta = 16$ Each Active Coordinate Moves One Factor





One active z_j from -2.5 to 2.5 around the code of one sprite; label: the factor with the largest $I(\mu_j; v_k)/H(v_k)$, and that value. $\beta = 16$: one row moves the sprite horizontally, one vertically, one scales it. VAE: position and size change together, some decodings show two faint sprites.

Published: a β-VAE Turns Heads Where a VAE Mixes Factors

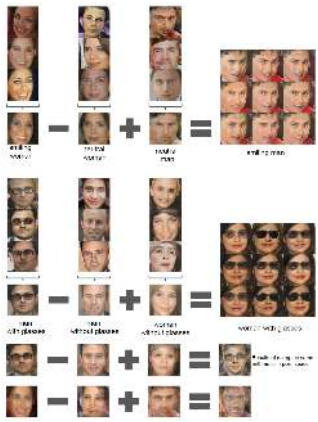


CelebA at 64×64; block: one coordinate from -3 to 3; row: one seed face

- β-VAE ($\beta = 250$): azimuth (left/right), smile, fringe, each in one coordinate
- VAE ($\beta = 1$): azimuth mixed with emotion, glasses, gender
- InfoGAN: a GAN-based method
- sunglasses were not isolated

Fig. 1 of I. Higgins et al., *β-VAE: Learning Basic Visual Concepts with a Constrained Variational Framework*, ICLR 2017, OpenReview; CelebA: Z. Liu, P. Luo, X. Wang, X. Tang, ICCV 2015, doi:10.1109/ICCV.2015.425

Published: Latent Arithmetic, "Man with Glasses − Man + Woman"



- column: latent vectors of 3 faces, averaged
- $\bar{\mathbf{z}}_{\text{man, glasses}} - \bar{\mathbf{z}}_{\text{man}} + \bar{\mathbf{z}}_{\text{woman}}$, decoded: a woman with glasses
- bottom rows: the same on pixels, misaligned overlaps
- a **GAN** (generative adversarial network), not a VAE, on web faces, not CelebA

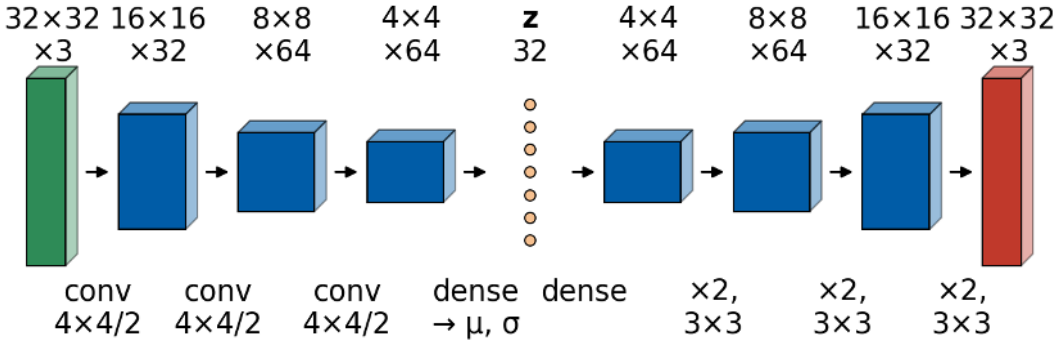
Figure 7 of A. Radford, L. Metz, S. Chintala, *Unsupervised Representation Learning with Deep Convolutional Generative Adversarial Networks*, arXiv:1511.06434

Published: A “Turn” Vector Rotates Faces from Left to Right



Latent difference between faces looking left and faces looking right (four of each, averaged), added in steps to the latent vectors of random generated faces (rows): the pose changes, the person largely stays. The generator is the deep convolutional GAN of Radford, Metz and Chintala. Figure 8 of A. Radford, L. Metz, S. Chintala, arXiv:1511.06434

Our CelebA β-VAE: 32×32 Faces on a CPU



- CelebA (CelebFaces Attributes): aligned celebrity photos, 40 binary labels; 20 000 + 2000 faces, 32×32
- 4×4 convolutions, stride 2 (Higgins et al.); $d = 32$, $\beta = 3$; 37 min, 8 CPU cores

Reconstructions Keep the Global Appearance, Not the Details



top: test faces $\mathbf{x}$; bottom: $\sigma(G_\theta(\boldsymbol{\mu}_\phi(\mathbf{x})))$

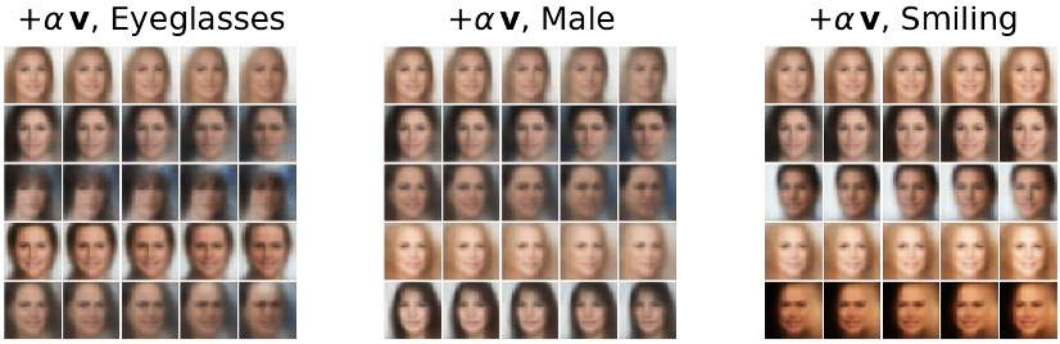
- test rate $R = 15.8$ nats per face; 18 of 32 coordinates active
- kept: pose, skin and hair colour, background, shape of the face
- lost: eyes, mouth, glasses; 15.8 nats is a small budget for a 32×32 colour image, blur is the price

Our Faces Model Separates Light, Skin and Hair, Not Pose



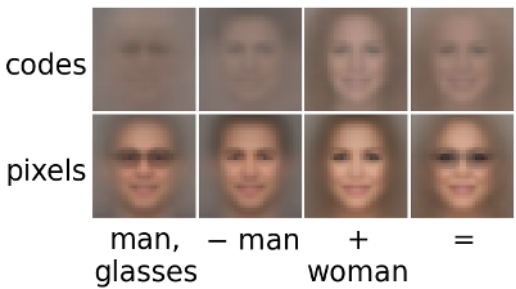
Two test faces (left, right block); row: one z_k from -3 to 3, 8 largest $\overline{\mathrm{KL}}_k$. Label: attribute a with the largest $|\rho(\mu_k, a)|$ over the training faces, ρ = Pearson correlation of μ_k with the 0/1 label. z_{10}: background; z_{32}: direction of the light; z_{25}: skin and hair colour; z_{23}: short-haired man $\rightarrow$ long-haired woman; z_2: hair length. No coordinate turns the head cleanly; $|\rho| \leq 0.35$: the labels describe details, the coordinates global appearance.

Attribute Directions Exist but Are Faint at 32×32



$\mathbf{v}_a = \langle \boldsymbol{\mu}_\phi(\mathbf{x}) \rangle_{a=1} - \langle \boldsymbol{\mu}_\phi(\mathbf{x}) \rangle_{a=0}$ (labels used only after training); rows: test faces without a, decoded $\boldsymbol{\mu}_\phi(\mathbf{x}) + \alpha\, \mathbf{v}_a$, $\alpha = 0, 0.5, 1, 1.5, 2$. Glasses: a dark band at the eyes for $\alpha = 2$; male: darker, broader face, shorter hair; smiling: brighter lower face.

Averaging Codes Gives a Mean Face; Aligned Pixels Can Be Averaged



Top: decoded $\bar{\boldsymbol{\mu}}_{\text{man, glasses}}$, $\bar{\boldsymbol{\mu}}_{\text{man}}$, $\bar{\boldsymbol{\mu}}_{\text{woman}}$ and $\bar{\boldsymbol{\mu}}_{\text{man, glasses}} - \bar{\boldsymbol{\mu}}_{\text{man}} + \bar{\boldsymbol{\mu}}_{\text{woman}}$ (training faces); bottom: the same with average images.

- mean codes lie near the centre: blurred "mean faces", faint glasses
- CelebA is *aligned*: eyes at the same pixels, so pixel arithmetic works here, unlike for the unaligned faces of Radford et al.

Disentangling Cannot Be Guaranteed Without Assumptions

If $\mathbf{z} \sim \mathcal{N}(0, I)$, then $R\mathbf{z} \sim \mathcal{N}(0, I)$ for every rotation R: the decoders $G(\mathbf{z})$ and $G(R^\top \mathbf{z})$ generate the *same* images; one may be disentangled, the other mixes all factors. The data cannot tell.

- Locatello et al.: for any factorised prior, infinitely many such reparametrisations exist $\Rightarrow$ unsupervised disentangling is impossible without *inductive biases* or supervision
- in practice the symmetry is broken by the *diagonal* encoder covariance, the architecture, the data, β, the seed
- their study: seeds and hyperparameters mattered more than the model; good models not selectable without true factors
- every factor found must be *identified afterwards*: traversals, or comparison with known quantities

F. Locatello, S. Bauer, M. Lucic, G. Rätsch, S. Gelly, B. Schölkopf, O. Bachem, ICML 2019, PMLR 97, 4114

8

Physics Learned by the β-VAE

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From Generating to Understanding

So far a VAE was judged by what it draws. Now we use it to learn about the *system*: the network sees only snapshots $\mathbf{s}$, no T, no phase, no Hamiltonian, and compresses each into a few numbers that β pushes to be independent. If the compression is good, these numbers are the quantities in which snapshots differ most: the macroscopic ones.

- Lecture 4: class activation maps showed *where* a classifier looks
- Lecture 5: the first principal component of Ising snapshots is the magnetisation (Wang 2016)
- a VAE latent variable correlates with the magnetisation (Wetzel 2017)
- a β-VAE cost function recovers spring constant and damping of a pendulum from its trajectory (Iten et al. 2020)

Known model: a check. Unknown model: a discovery tool. L. Wang, Phys. Rev. B **94**, 195105 (2016), doi; S. J. Wetzel, Phys. Rev. E **96**, 022140 (2017), doi; R. Iten, T. Metger, H. Wilming, L. del Rio, R. Renner, Phys. Rev. Lett. **124**, 010508 (2020), doi

The Probing Protocol

1. train a β-VAE on snapshots only (bits $b_i = (s_i + 1)/2$, Bernoulli decoder), $d = 8$, $\beta \in \{1, 4\}$
2. keep the **active** coordinates: $\overline{\mathrm{KL}}_k = \left\langle \frac{1}{2}(\mu_k^2 + \sigma_k^2 - \ln \sigma_k^2 - 1) \right\rangle_{\mathbf{s}} > 0.1$ nats on held-out snapshots
3. **probe**: latent mean $\mu_k(\mathbf{s})$ of every held-out snapshot, Pearson correlation with an observable $O(\mathbf{s})$,

$$\rho_{kO} = \frac{\langle \mu_k O \rangle - \langle \mu_k \rangle \langle O \rangle}{\sqrt{\mathrm{Var}\, \mu_k \; \mathrm{Var}\, O}} \in [-1, 1]$$

   ± 1: exact linear relation, 0: no linear relation
4. table (coordinates $\times$ observables) and scatter plots of μ_k against the best-matching O

Observables Computed from the Spins

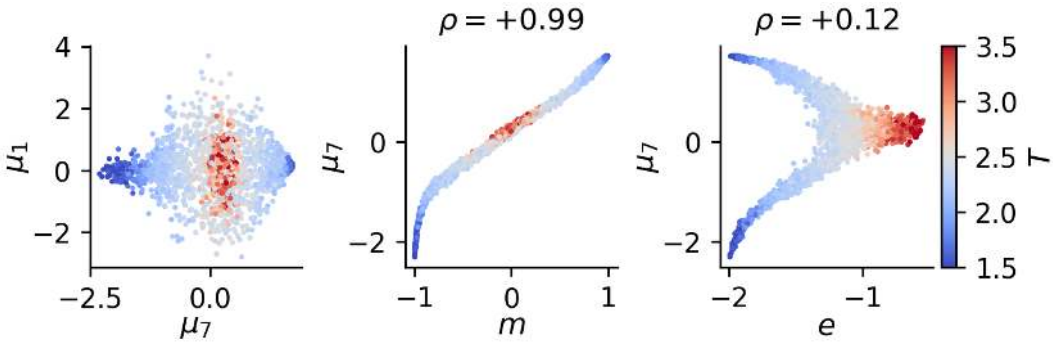
$N = L^2$ spins $s_{x,y} = \pm 1$:

- magnetisation $m = \frac{1}{N} \sum_{x,y} s_{x,y}$ and $|m|$: ferromagnetic order
- energy per spin $e = H/N$: satisfied and broken bonds
- stripe amplitudes $m_x = \frac{1}{N} \sum_{x,y} (-1)^x s_{x,y}$, $m_y = \frac{1}{N} \sum_{x,y} (-1)^y s_{x,y}$: spins alternating along x or y; $|m_x|$, $|m_y|$, $m_{\mathrm{str}} = \sqrt{m_x^2 + m_y^2}$
- simulation parameters: temperature T, coupling ratio $g = J_2/J_1$

ρ detects *linear* relations: a coordinate $\mu_k \propto m$ gives $|\rho| \approx 1$ with m but $\rho \approx 0$ with $|m|$; a symmetric relation such as $\mu_k \approx m^2$ needs $|m|$ or $|\mu_k|$.

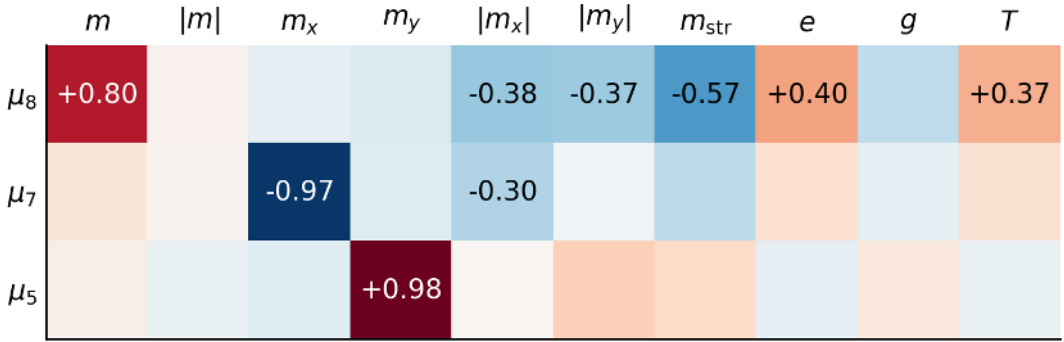
β below is always the KL weight, not $1/T$.

One Coordinate Is the Magnetisation



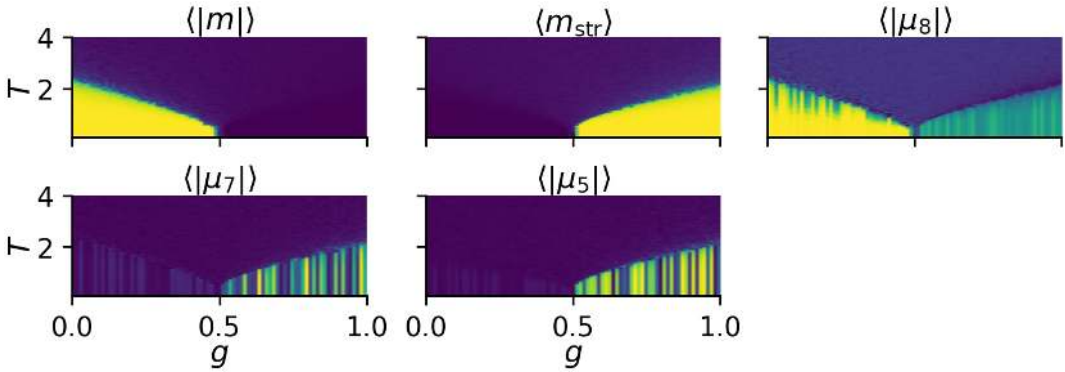
2D Ising, $L = 30$, $T \in [1.5, 3.5]$, 9472 snapshots (20% held out, shown, colour $= T$); $\beta = 4$, $d = 8$. Left: two most active coordinates; middle, right: the coordinate with the largest $|\rho|$ with m, with e. $\rho(\mu_7, m) = +0.99$ ($\beta = 1$ and 4); all other coordinates $|\rho| \leq 0.16$ with $m, |m|, e, T$. Ordered phases at the two ends of the μ_7 axis, disordered phase at its centre; e is related to $|\mu_7|$ ($\rho(|\mu_7|, e) = -0.88$), invisible to a linear probe.

J_1–J_2: Three Coordinates Are m, m_x and m_y



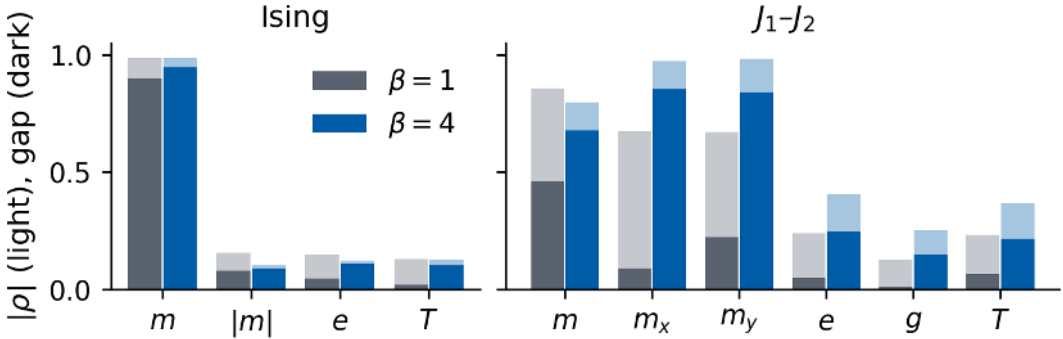
J_1–J_2, $L = 32$, 81×79 grid of (g, T), 12 snapshots per point (10 training, 2 held out); $\beta = 4$; rows: active coordinates, colour and numbers: $\rho(\mu_k, O)$ on held-out snapshots (numbers if $|\rho| \geq 0.3$). $\rho(\mu_7, m_x) = -0.97$, $\rho(\mu_5, m_y) = +0.98$, $\rho(\mu_8, m) = +0.80$, found without g, T or H. With $\beta = 1$: m in one coordinate (-0.86), m_x, m_y mixed over two (a rotated stripe plane).

The Coordinates Map the Phase Diagram



Over (g, T): mean over 12 snapshots of $|m|$, m_{str}, $|\mu_k|$ ($\beta = 4$); colour: min (dark) to max (yellow). $|\mu_8|$: ferromagnet, with some leakage into the stripe phase; $|\mu_7|$, $|\mu_5|$: stripe phase, in complementary bands (an artefact of the annealed chains; in equilibrium both stripe directions are equally likely); phase boundaries as in $\langle|m|\rangle$, $\langle m_{\mathrm{str}}\rangle$.

Larger β Gives Each Order Parameter Its Own Coordinate



For each observable O, over the active coordinates: light bar $|\rho|_{(1)} = \max_k |\rho_{kO}|$; dark bar: gap $|\rho|_{(1)} - |\rho|_{(2)}$ to the second largest (the MIG idea with correlations): large gap = one coordinate, and only one, follows O. Ising: one order parameter, gap 0.90 ($\beta = 1$) and 0.95 ($\beta = 4$). J_1–J_2: gap for m_x 0.09 $\to$ 0.85, m_y 0.22 $\to$ 0.84, m 0.46 $\to$ 0.68; e, g, T: no linear match ($|\rho| \leq 0.40$).

From a Check to a Discovery Tool

On an unknown model, an active coordinate whose $\langle|\mu_k|\rangle$ changes sharply across a parameter range is a *candidate order parameter*. Next step: express it through the spins.

Example on J_1–J_2, as if the answer were unknown: least-squares fit of μ_k by 8 standardised features $m, m_x, m_y, |m|, |m_x|, |m_y|, c_{\mathrm{nn}}, c_{\mathrm{nnn}}$ ($c_{\mathrm{nn}} = \frac{1}{4N}\sum_i s_i \sum_{j\in \mathrm{nn}(i)} s_j$, c_{nnn} the same for diagonal neighbours):

$$R^2 = 1 - \frac{\sum_{\mathbf{s}} (\mu_k - \hat{\mu}_k)^2}{\sum_{\mathbf{s}} (\mu_k - \langle \mu_k \rangle)^2} \quad \text{(fraction of the coordinate explained by the features)}$$

$\beta = 4$ model: $R^2 = 0.982$ for z_8 (largest coefficient m, then $-|m_x|$), 0.992 for z_7 (m_x), 0.989 for z_5 (m_y).

- the largest coefficient names the coordinate; a low R^2 means a quantity outside our library
- richer libraries and symbolic regression turn a coordinate into a formula

What a Probe Cannot Tell

- correlation **confirms, does not explain**: m, e and T are themselves correlated across a transition
- a coordinate may **mix** quantities; sign and scale of a coordinate are arbitrary
- a **nonlinear** relation ($\mu_k \approx m^2$) escapes a linear probe of m
- which coordinates appear depends on the model, the data, the architecture, β and the seed (Locatello et al.)
- **we only find what we probe for**: a coordinate matching none of our observables is noise or a quantity we have not thought of

Examples in Lab 026, held-out snapshots

Ising, $\beta = 4$: $\rho(|\mu_7|, \cdot) = +0.92$ ($|m|$), -0.88 (e), -0.78 (T); linear: $\rho(\mu_7, m) = +0.99$, $\rho(\mu_7, |m|) = -0.10$. J_1–J_2, $\beta = 1$: μ_5 mixes m_x (-0.58) and m_y ($+0.67$).

Conditioning and Disentangling Answer Different Questions

	conditional VAE (Labs 024, 025)	β-**VAE** (Lab 026)
training data	pairs $(\mathbf{x}, \mathbf{c})$	images $\mathbf{x}$ only
loss	$D + R$, $\mathbf{c}$ into encoder and decoder	$D + \beta R$
factor	chosen by us, given as $\mathbf{c}$	found by the network, named afterwards
guarantee	$\mathbf{c}$ controls what it labels	none in general
$\mathbf{z}$ carries	all but $\mathbf{c}$	all, ideally one factor per coordinate
sampling	fix $\mathbf{c}$, draw $\mathbf{z}$; or one $\mathbf{z}$, many $\mathbf{c}$	draw $\mathbf{z}$; move one found z_k
question	"generate this"	"what varies in these data?"

Complements: factors found by a β-VAE can become labels of a cVAE; physics: condition on (g, T) to generate, an unconditioned β-VAE to find collective variables.

9

Physical Parameters from Trajectories



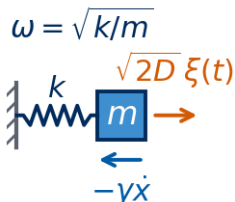
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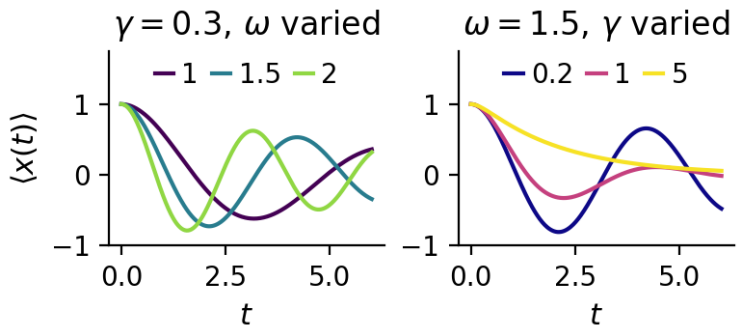
Frequency and Damping Shape a Noisy Oscillation

$$\ddot{x} = -\omega^2 x - \gamma\,\dot{x} + \sqrt{2D}\,\xi(t), \qquad \langle \xi(t) \rangle = 0, \quad \langle \xi(t)\,\xi(t') \rangle = \delta(t - t')$$

$\omega = \sqrt{k/m}$: frequency without friction; γ: friction rate; ξ: Gaussian white noise (independent kicks); D: noise strength ($= \gamma k_B T/m$ for a heat bath)

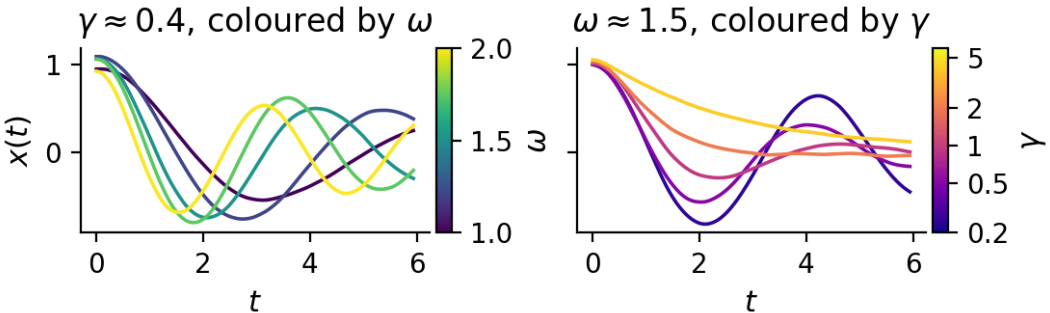
underdamped $\gamma < 2\omega$: oscillation at $\omega_d = \sqrt{\omega^2 - \gamma^2/4}$ inside $e^{-\gamma t/2}$; **overdamped** $\gamma > 2\omega$: creep back at the slow rate $\approx \omega^2/\gamma$





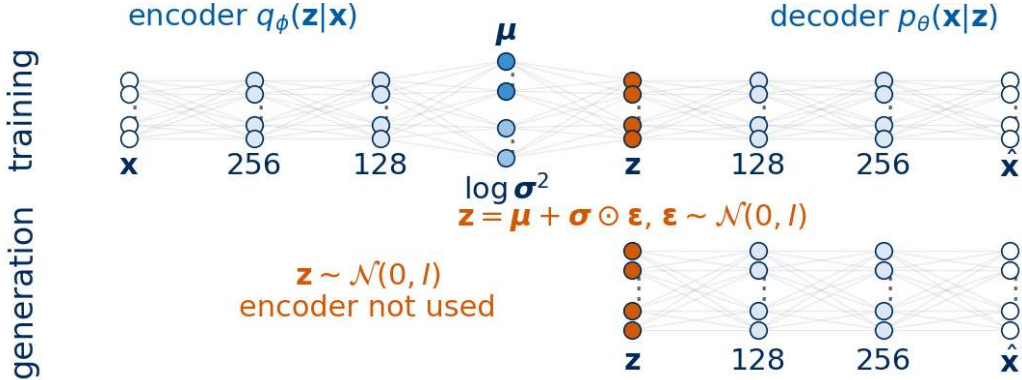
Simulated Trajectories with Known Parameters

Euler–Maruyama, $\Delta t = 0.01$, independent $\eta_n \sim \mathcal{N}(0,1)$:
$x_{n+1} = x_n + v_n \Delta t, \quad v_{n+1} = v_n + (-\omega^2 x_n - \gamma v_n)\Delta t + \sqrt{2D\Delta t}\,\eta_n$
noise step $\propto \sqrt{\Delta t}$ since $\int_t^{t+\Delta t} \xi\, ds$ has variance Δt; check: $\langle x^2 \rangle = D/(\gamma\omega^2) = 0.00178$, recursion 0.00186, 20 000 runs 0.00185
20 000 trajectories, 100 points, $t \in [0, 5.94]$: $\omega \in [1, 2]$, $\ln \gamma$ uniform in $[\ln 0.2, \ln 6]$ (21% overdamped); nuisance $x_0 \in [0.5, 1.5]$, $v_0 \in [-0.5, 0.5]$; $D = 0.002$

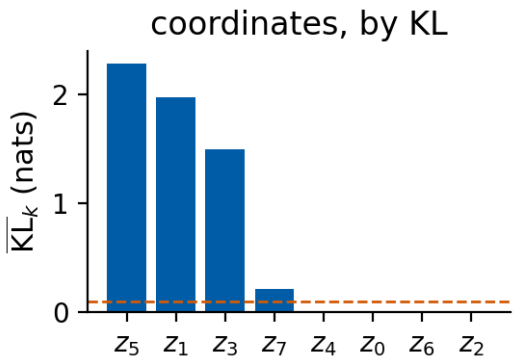


A β-VAE Compresses Each Trajectory

Gaussian decoder $p_\theta(\mathbf{x}|\mathbf{z}) = \prod_i \mathcal{N}(x_i;\, G_\theta(\mathbf{z})_i,\, \sigma_x^2) \Rightarrow$
$\mathcal{L}_\beta = \mathbb{E}_{q_\phi}\left[\|\mathbf{x} - G_\theta(\mathbf{z})\|^2/(2\sigma_x^2)\right] + \beta\,\mathrm{KL}(q_\phi(\mathbf{z}|\mathbf{x})\,\|\,\mathcal{N}(0, I))$
reconstruction = mean squared error $\times\, 100/(2\sigma_x^2)$, $\sigma_x = 0.1$; G_θ: mean trajectory; $d = 8$;
$\beta \in \{0.5, 1, 4, 8\}$, 3 seeds each



Damping Gets Its Own Coordinate, Frequency Does Not



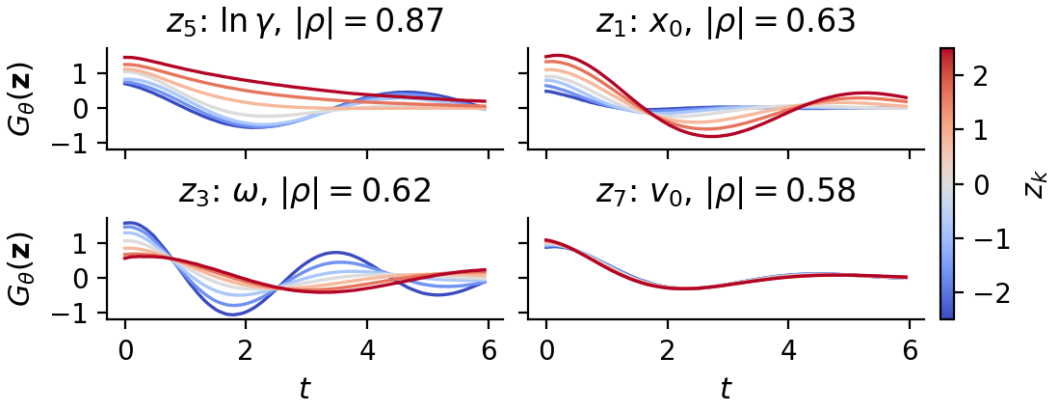
$|\rho(\mu_k, \text{factor})|$

	ω	$\ln\gamma$	x_0	v_0
z_5	0.27	0.87	0.19	0.18
z_1	0.40	0.30	0.63	0.26
z_3	0.62	0.11	0.46	0.11
z_7	0.08	0.16	0.23	0.58

$\beta = 4$, seed 2 (largest MIG of three seeds), 5000 test trajectories; ρ: Pearson correlation of latent mean μ_k and factor; active: $\overline{\mathrm{KL}}_k > 0.1$ nats (dashed)

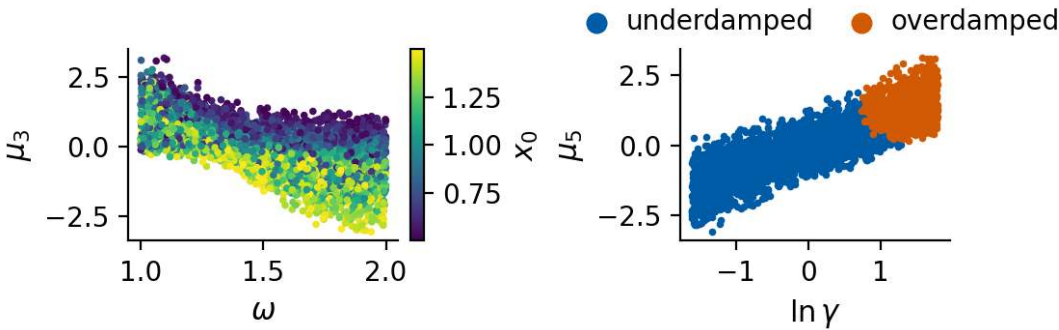
4 active coordinates for 4 factors; damping $|\rho| = 0.87$, clean; ω (0.62) and x_0 (0.63) shared by two coordinates

Traversals Separate Damping from Frequency



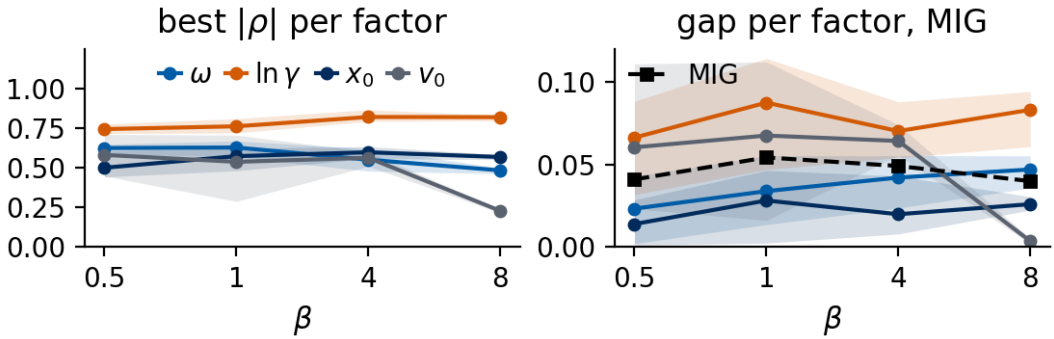
$\boldsymbol{\mu}$ of one test trajectory ($\omega \approx 1.5$, $\gamma \approx 1$, $x_0 \approx 1$, $v_0 \approx 0$); one z_k from -2.5 to 2.5, the others fixed; title: factor with the largest $|\rho|$. The ω-coordinate also moves $x(0)$.

Overdamped: the Damping Coordinate Measures ω^2/γ



- right: $|\rho(\mu_\gamma, \ln\gamma)| = 0.80$ underdamped, 0.13 overdamped, where $|\rho(\mu_\gamma, \ln\frac{\omega^2}{\gamma})| = 0.69$: the slow rate is what the data show
- left: slope of μ_ω grows with x_0; R^2 of a fit: ω 0.38; $\omega, x_0, \omega x_0$ 0.63

Disentangling Stays Partial for Every β



mean over 3 seeds (band: min–max); $\mathrm{MIG} = \frac{1}{K}\sum_k [I(\mu_{j_1}; v_k) - I(\mu_{j_2}; v_k)]/H(v_k)$ (β-VAE section), 20 bins, active coordinates

MIG 0.041, 0.054, 0.049, 0.040 for $\beta = 0.5, 1, 4, 8$; true factors 0.92, ω, x_0 rotated by 45° 0.46

active coordinates per seed: 5, 5, 5; 4, 4, 4; 5, 4, 4; 3, 3, 3: $\beta = 8$ drops v_0

The Most Visible Parameter Disentangles

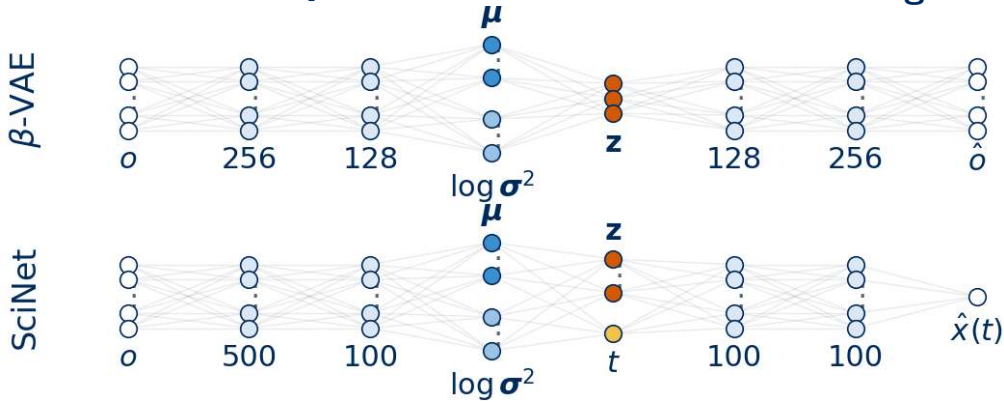
Fisher information of the Gaussian likelihood, parameters $\mathbf{p} = (\omega, \ln\gamma, x_0, v_0)$:

$$F_{ab} = \frac{1}{\sigma_x^2} \sum_i \frac{\partial\langle x(t_i)\rangle}{\partial p_a} \frac{\partial\langle x(t_i)\rangle}{\partial p_b}$$

- **visibility** $\sqrt{F_{aa}}\,\mathrm{std}(p_a)$: how many σ_x the trajectory moves for a typical change of p_a; $(\omega, \ln\gamma, x_0, v_0)$: 17, 23, 12, 5: γ most visible, v_0 least, dropped first
- **confusion** $|F_{ab}|/\sqrt{F_{aa}F_{bb}} = |\cos|$ of sensitivity vectors: ω–v_0 0.79, ω–$\ln\gamma$ 0.69, ω–x_0 only 0.24: that mixing is nonlinear (sensitivity to $\omega \propto$ amplitude)
- many periods in the window, or x_0 of both signs, fold the map $\mathbf{p} \mapsto \mathbf{x}$: hence narrow ranges, $x_0 > 0$

medians over 2000 test parameter points; derivatives of $e^{At}\mathbf{s}(0)$ with `jax.jacfwd`

SciNet Answers Questions Instead of Reconstructing



Iten et al.: $o = 50$ positions on $[0, 5]$ of $m\ddot{x} = -\kappa x - b\dot{x}$, no noise, amplitude and phase fixed; question $t \in [0, 10]$; cost $(\hat{x}(t) - x(t))^2 + \beta\,\mathrm{KL}$, $\beta = 10^{-3}$, 3 latent neurons. Reported: one neuron $\approx$ linear in κ, one in b, the third unused.

R. Iten et al., Phys. Rev. Lett. **124**, 010508 (2020), doi:10.1103/PhysRevLett.124.010508; arXiv:1807.10300 (Box 1, Tables I–II, Fig. 6)

SciNet Needs 1000 Epochs to Isolate κ and b

Same data; SciNet 100, β-VAE 60 epochs unless stated; best $|\rho|$ / second best over active neurons (one neuron per parameter: high / low)

	SciNet, $\beta = 10^{-3}$		β-VAE, $\beta = 4$	
data	κ	b	κ	b
paper setting	0.96 / 0.85	0.50 / 0.21	0.99 / 0.01	0.97 / 0.08
paper setting, 1000 epochs	1.00 / 0.05	0.98 / 0.06		
+ noise	0.98 / 0.84	0.49 / 0.10	0.98 / 0.03	0.96 / 0.08
+ noise, random x_0, v_0	0.91 / 0.43	0.30 / 0.02	0.90 / 0.06	0.79 / 0.26

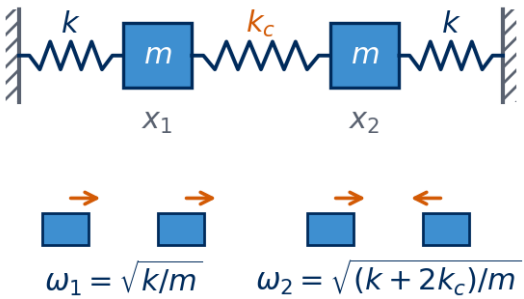
- 1000 epochs as in the paper: κ, b separate, third neuron off, rms 0.027 (paper < 0.02)
- KL weight per squared error: β-VAE $4/50 = 0.08$, SciNet 10^{-3}: SciNet's pull to the prior is $80\times$ weaker
- random x_0, v_0: SciNet stores κ, x_0, loses b; β-VAE keeps κ, b, x_0, drops v_0

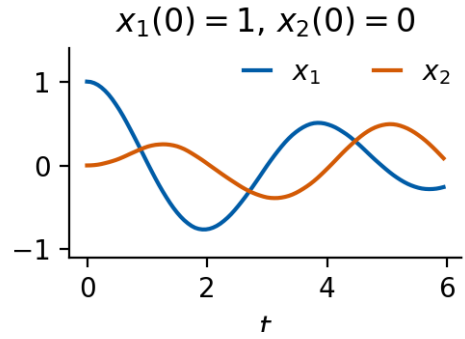
Two Coupled Masses Are Two Independent Normal Modes

$$\ddot{x}_{1,2} = -k\,x_{1,2} - k_c\,(x_{1,2} - x_{2,1}) - \gamma\,\dot{x}_{1,2} + \sqrt{2D}\,\xi_{1,2}(t), \qquad m = 1$$

$q_\pm = (x_1 \pm x_2)/\sqrt{2}$: two independent noisy oscillators with $\omega_1 = \sqrt{k}$, $\omega_2 = \sqrt{k + 2k_c}$; the springs enter the trajectories *only* through ω_1, ω_2.

$k \in [1, 4]$, $k_c \in [0.2, 2]$, $\ln\gamma$ uniform in $[\ln 0.2, \ln 2]$; does a coordinate follow ω_2 or k_c? In the data $\rho(\omega_2, k_c) = 0.75$, $\rho(\omega_2, k) = 0.63$





Coupled Masses: Neither Springs Nor Normal Modes

		z_1	z_8	z_4	z_0	z_9	z_6	z_5
springs	k	0.68	0.16	0.26	0.22	0.32	0.17	0.08
	k_c	0.18	0.07	0.12	0.65	0.12	0.26	0.09
	$\ln\gamma$	0.47	0.08	0.48	0.22	0.47	0.11	0.09
modes	ω_1	0.69	0.17	0.24	0.21	0.32	0.17	0.09
	ω_2	0.60	0.03	0.05	0.63	0.12	0.09	0.13
	$\ln\gamma$	0.47	0.08	0.48	0.22	0.47	0.11	0.09

input $(x_1(t_i), x_2(t_i))$, 200 numbers; $d = 10$, $\beta = 4$, seed 1; $|\rho(\mu_k, \cdot)|$, active coordinates

- one coordinate follows k (0.68), i.e. $\omega_1 = \sqrt{k}$ (0.69); one follows k_c (0.65) and ω_2 (0.63) equally: a pure ω_2 or k_c coordinate would give 0.75 with the other
- friction spread over three coordinates ($|\rho| \leq 0.48$); initial conditions take four

From Physical Parameters to Diffusion Models

- a β-VAE trained on trajectories alone finds *how many* numbers describe them (4 for 4 factors at $\beta = 1$) and isolates the most visible parameter (damping); weakly visible or mutually confusable ones stay mixed
- check against physics before naming a coordinate: ω^2/γ in the overdamped regime is a correct answer, not a failure
- SciNet asks questions about the system instead of reconstructing the input; trained as long as in the paper it isolates κ and b, as the β-VAE does with a stronger KL term
- **Lecture 7**: the Langevin equation $\dot{\mathbf{x}} = -\nabla U(\mathbf{x}) + \sqrt{2D}\,\boldsymbol{\xi}$, integrated with Euler–Maruyama as here, samples $p \propto e^{-U/D}$; diffusion models learn the score $\nabla \log p$ and use Langevin-type dynamics to turn noise into data

R. Iten et al., PRL **124**, 010508 (2020), doi:10.1103/PhysRevLett.124.010508; P. E. Kloeden, E. Platen, Springer (1992), doi:10.1007/978-3-662-12616-5

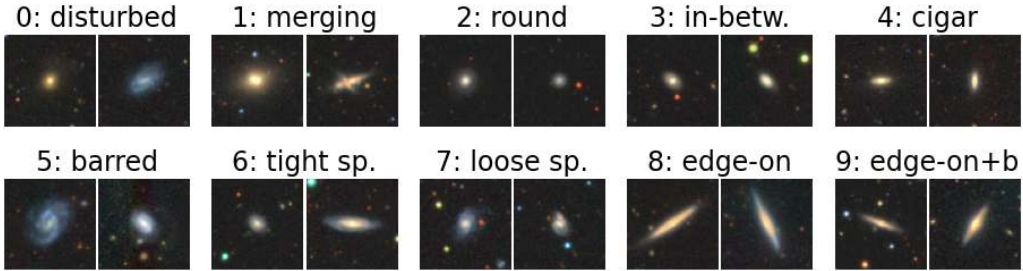
10

Galaxy Morphology with a β-VAE



JAGIELLONIAN UNIVERSITY
IN KRAKOW

Galaxy10 DECaLS: 17 736 Galaxies in Ten Classes



- one sample: 64×64 pixels of $1.05''$; filter z (near infrared) $\rightarrow$ red, r (red) $\rightarrow$ green, g (blue-green) $\rightarrow$ blue; value: arcsinh-stretched brightness
- images: Dark Energy Camera Legacy Survey (DECaLS); classes: Galaxy Zoo votes

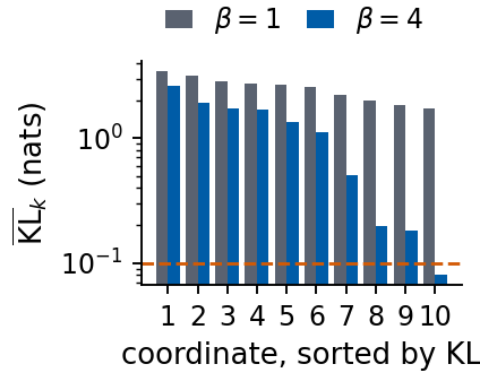
H. W. Leung, J. Bovy, Galaxy10 DECaLS, CC BY 4.0, Zenodo; M. Walmsley et al., MNRAS 509, 3966 (2022); A. Dey et al., AJ 157, 168 (2019)

The β-VAE Loss with a Gaussian Decoder

Galaxy pixels are continuous, so a Gaussian replaces the BCE of the digit model:

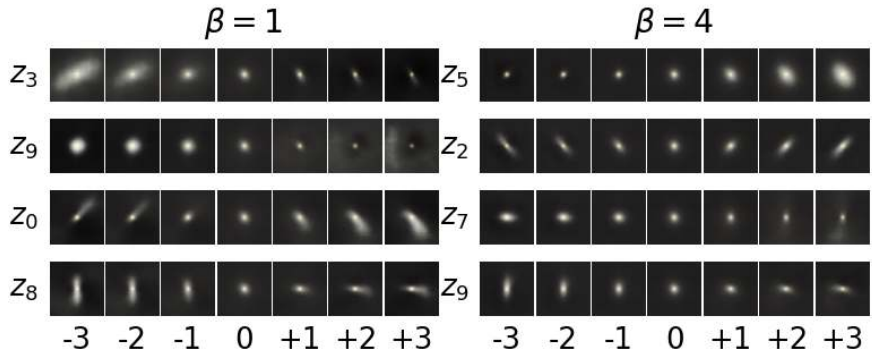
$$\mathcal{L}_\beta(\mathbf{x}) = \mathbb{E}_{q_\phi}\left[\frac{\|\mathbf{x} - \hat{\mathbf{x}}_\theta(\mathbf{z})\|^2}{2\sigma_x^2}\right] + \beta \sum_k \mathrm{KL}_k$$

- $\mathbf{x} \in [0,1]^{12288}$: image; $\hat{\mathbf{x}}_\theta(\mathbf{z})$: decoded image; $\sigma_x = 0.15$
- $q_\phi(\mathbf{z}|\mathbf{x}) = \mathcal{N}(\boldsymbol{\mu}, \mathrm{diag}\,\boldsymbol{\sigma}^2)$; prior $\mathcal{N}(0, I)$
- $\mathrm{KL}_k = \frac{1}{2}(\mu_k^2 + \sigma_k^2 - \ln\sigma_k^2 - 1)$: nats in coordinate k; β: price per nat; $\beta = 1$: $\mathcal{L}_1 = -\mathrm{ELBO}$



$\overline{\mathrm{KL}_k}$: mean over 2 736 test images; per image 25.3 $\rightarrow$ 11.4 nats; active: 10 $\rightarrow$ 9

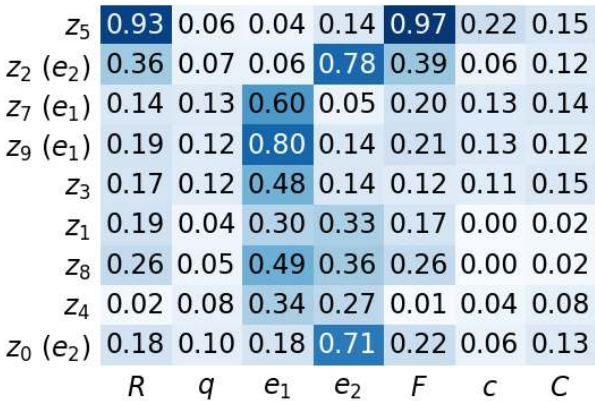
Traversals Separate Size, Brightness and Shape



Row: $\hat{\mathbf{x}}_\theta(\mathbf{z})$, $z_k = \mu_k(\mathbf{x}_0) + v$, other coordinates fixed

- $\beta = 4$: z_5 size and brightness; z_2 diagonal stretch; z_7, z_9 stretch along the axes
- $\beta = 1$: size, brightness and shape change together; no row changes the colour

Latent Coordinates Measured Against Physical Quantities



$\beta = 4$; $|\rho|$: Spearman rank correlation of $\mu_k(\mathbf{x})$ and a quantity, 2 736 test galaxies

From the pixels alone:

- R: radius of the mask
- $q = \sqrt{\lambda_2/\lambda_1}$: axis ratio, $\lambda_{1,2}$ second moments; disk: $q \approx \cos i$
- $(e_1, e_2) = \varepsilon\,(\cos 2\theta, \sin 2\theta)$, $\varepsilon = \frac{1-q^2}{1+q^2}$, θ: orientation
- F: brightness; c: red $-$ blue
- $C = r_{80}/r_{20}$: concentration

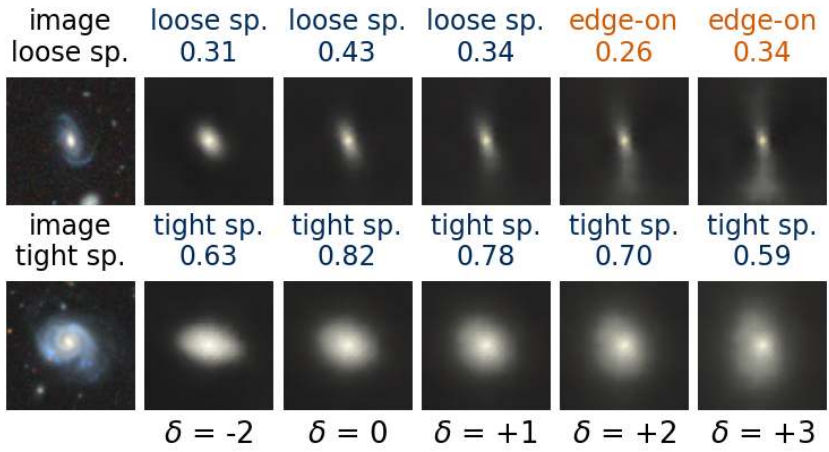
name: $|\rho| \geq 0.5$, 0.15 above the row

The Shape Is Stored as Two Ellipticity Components

- size and brightness: one coordinate at $\beta = 4$, z_5 ($|\rho| = 0.93$ with R, 0.97 with F); spread over five coordinates at $\beta = 1$
- orientation: z_9, z_7 follow e_1 (0.80, 0.60); z_2, z_0 follow e_2 (0.78, 0.71)
- axis ratio: $|\rho| \leq 0.22$ for every coordinate: both signs of e_1 make a galaxy longer, so q is not monotonic in any coordinate; inclination = length of (e_1, e_2)
- colour: $|\rho| \leq 0.27$, the reconstructions are grey
- correlation gap per quantity = largest − second-largest $|\rho|$ over coordinates; mean over quantities: 0.12 ($\beta = 1$), 0.22 ($\beta = 4$), 0.14 ($\beta = 4$, second seed)
- the seed changes the organisation as much as β: unsupervised disentangling is not identifiable without inductive biases

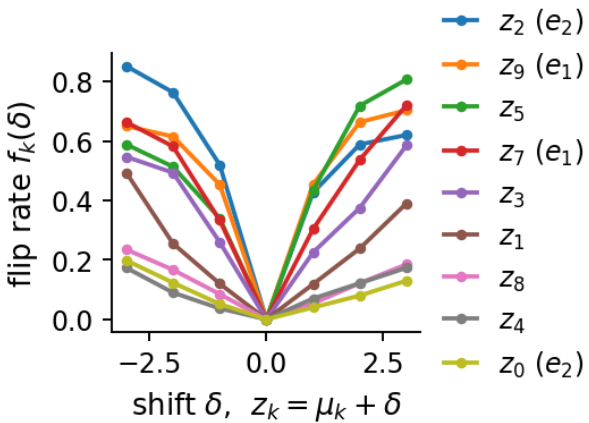
F. Locatello et al., ICML 2019, arXiv:1811.12359

Counterfactual Galaxies: Elongation Turns Spirals Edge-On



$\mathbf{z} = \boldsymbol{\mu}_\phi(\mathbf{x}) + \delta\, \mathbf{e}_7$, $\delta > 0$: smaller q; of 249 spirals 12% edge-on at $\delta = 3$, 6% at -3

The Classifier Responds Most to Shape and Size



Convolutional neural network (CNN): accuracy 0.640, on reconstructions 0.315. $f_k(\delta)$: fraction of 600 test galaxies whose class changes when $z_k = \mu_k + \delta$

- $|\delta| = 2$: 68% for z_2 (e_2), 10% for z_0 (e_2, barely used by the decoder)
- flips: round $\rightarrow$ loose spiral, edge-on
- decoded grey blobs lie outside the training data

Summary: Learning and Sampling a Distribution

- an **autoencoder** maps each image to a point of a latent space but learns no distribution on it: islands of codes, an empty sea that decodes to non-digits, random latent vectors give no digits
- a **latent-variable model** $p_\theta(\mathbf{x}) = \int p(\mathbf{z})\, p_\theta(\mathbf{x}|\mathbf{z})\, d\mathbf{z}$ is sampled in two steps, first $\mathbf{z}$, then $\mathbf{x}$; fitting it by maximum likelihood minimises $\mathrm{KL}(p \,\|\, p_\theta)$
- a **VAE** trains decoder and encoder clouds $q_\phi(\mathbf{z}|\mathbf{x})$ on $\mathrm{ELBO} = \mathbb{E}_q \log p_\theta(\mathbf{x}|\mathbf{z}) - \mathrm{KL}(q \,\|\, p) \le \log p_\theta(\mathbf{x})$: reconstruction (BCE) plus the price of information, which pulls all clouds onto $\mathcal{N}(0, I)$
- **sampling**: keep prior and decoder; $\mathbf{z} \sim \mathcal{N}(0, I)$, decode; the decoder mean is blurry, a sample is speckled (pixels independent given $\mathbf{z}$); new combinations, not copies; prior holes make samples less clear than reconstructions; single coordinates mix several properties

Summary: Judging Samples and Conditional Generation

- a **classifier trained on real data** referees synthetic data, always against the real-data baseline
- a **conditional VAE** takes a known label $\mathbf{c}$: sampling at fixed $\mathbf{c}$ draws from $p_\theta(\mathbf{x}|\mathbf{c})$, $\mathbf{c}$ selects, $\mathbf{z}$ carries the style; unseen $\mathbf{c}$ are interpolated
- **spins**: generating at $T =$ draw $\mathbf{z}$, then every spin independently, a mean-field-like sampler; with the $\mathbb{Z}_2$ symmetry and a prior $p_\psi(\mathbf{z}|T)$ it reproduces the phase, the order parameter and the J_1–J_2 phase diagram at unseen (g, T), not the energy and its fluctuations: it is not a sample of $e^{-H/T}/Z$

Summary: Disentanglement and Physical Parameters

- a β-**VAE** minimises $D + \beta R$ (distortion + $\beta \times$ rate): β trades sharpness for a simpler, more independent code; it can *discover* factors that a cVAE must be given, but without labels disentangling is not guaranteed and every factor must be identified afterwards
- **physical parameters** from data: on noisy trajectories one coordinate follows the damping, the frequency stays mixed with the amplitude; on galaxies, size and orientation get coordinates, the axis ratio does not – disentangling is partial and depends on the data and the seed

Thank You!

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Website · GitHub · Google Scholar

